

Stability Criterion and Stability Enhancement for a Thruster-Assisted Underwater Hexapod Robot

Lepeng Chen , Rongxin Cui , *Member, IEEE*, Weisheng Yan , Chenguang Yang , *Fellow, IEEE*, Zhijun Li , *Fellow, IEEE*, Hui Xu , and Haitao Yu 

Abstract—The stability criterion is critical for the design of legged robots' motion planning and control algorithms. If these algorithms cannot theoretically ensure legged robots' stability, we need many trials to identify suitable parameters for stable locomotion. However, most existing stability criteria are tailored to robots driven solely by legs and cannot be applied to thruster-assisted legged robots. Here, we propose a stability criterion for a thruster-assisted underwater hexapod robot by finding maximum and minimum allowable thruster forces and comparing them with the current thrusts to check its stability. On this basis, we propose a method to increase the robot's stability margin by adjusting the value of thrusts. This process is called stability enhancement. The criterion uses the optimization method to transform multiple variables such as attitude, velocity, acceleration of the robot body, and the angle and angular velocity of leg joints into one kind of variable (thrust) to judge the stability directly. In addition, the stability enhancement method is straightforward to implement because it only needs to adjust the thrusts. These provide insights into how multiclass forces such as inertia force, fluid force, thrust, gravity, and buoyancy affect the robot's stability.

Index Terms—Stability criterion, stability enhancement, thruster-assisted underwater hexapod robot.

I. INTRODUCTION

STABILITY is a critical issue for underwater legged robots because they may suffer from multiclass disturbances while walking, such as waves, currents, towing force of cable, and uncertain leg-terrain interaction force [1], [2]. The robot is stable if it can recover to the preset motion state by itself under the disturbances. Otherwise, it is unstable. In addition, if motion planning and control algorithms cannot theoretically ensure stability, they cannot be applied to other robots directly, and

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Lepeng Chen, Rongxin Cui, Weisheng Yan, Hui Xu, and Haitao Yu are with the School of Marine Science and Technology, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: r.cui@nwpu.edu.cn).

Chenguang Yang is with the Department of Computer Science, University of Liverpool, L69 3BX Liverpool, U.K..

Zhijun Li is with the School of Mechanical Engineering, and Translational Research Center, Shanghai Yang Zhi Rehabilitation Hospital (Shanghai Sunshine Rehabilitation Center), Tongji University, Shanghai 200070, China.

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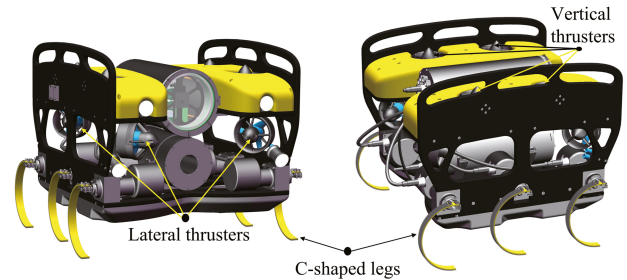


Fig. 1. Underwater hexapod robot has six C-shaped legs and eight thrusters (four lateral and four vertical thrusters). The robot body and legs are connected by hip joints [4], [5]. Legs and thrusters simultaneously drive the robot, which can walk on the inclined and rough underwater terrains.

many trials are needed to find the feasible parameters for stable locomotion [3]. Thus, a clear stability criterion is essential, one that not only evaluates the robot's stability but also supports the development of stability constraints for designing robust planning and control algorithms.

To perform some meticulous small-range operations on the underwater structure's surface, the robot should be able to walk on it. Then, we develop a thruster-assisted underwater hexapod robot (see Fig. 1) [4], [5]. Unlike underwater robots driven by legs that only walk on the seabed, our robot can walk on structures with various inclinations and surface roughness.

Most existing stability criteria are with robots only driven by legs, and they cannot be applied to our robot because it is driven by legs and thrusters simultaneously. Moreover, it is challenging to identify how the other multiclass forces, such as inertia force, fluid force, gravity, and buoyancy, affect the robot's stability and enhance it by regulating thrusts.

A. Related Works

The dynamic model can be used to predict the legged robot's motion by establishing the relationship between multiclass forces and the motion states. Knowing the future motions contributes to evaluating the robot's stability. The model mainly consists of two types: inverted pendulum model [6], [7], [8], [9] and force balancing model [10], [11], [12], [13]. The former is established by simplifying the robot as an inverted pendulum, and the latter is an equilibrium equation to balance multiclass forces, e.g., leg-terrain interaction forces, inertia forces, and other forces.

Inspired by the spring-loaded inverted pendulum (SLIP) model for ground legged robots, Calisti et al. [6] proposed a new dynamic model, called underwater spring-loaded inverted

pendulum (U-SLIP), for underwater legged robots. On this basis, Picardi et al. [7] presented a three-dimensional (3-D) U-SLIP model that includes an underwater legged robot's lateral dynamic motion. Lin et al. [8], [9] presented a rolling inverted pendulum model for a ground hexapod robot by using the Lagrange method and incorporating the rolling character caused by C-shaped legs. These elegant models can only describe the motion of walking on flat terrain and need an assumption that the leg-terrain friction should be large enough.

Several attractive force balancing models are developed in [10], [11], and [12] for ground quadruped robots, that relax the above flat terrain and sufficient friction assumptions. Calisti et al. [13] presented a force balancing model for a novel octopus-inspired underwater robot by considering leg-terrain interaction forces and drags arising from the sculling motion of the legs. These models apply only to ground robots with point-contact legs. Thus, we need to establish a force balancing model for our robot by considering its unique driving manner and fluid forces.

Several insightful stability criteria are presented for legged robots and can be divided into three categories: zero-tilting moment point (ZMP) criterion [14], [15], Poincare return map criterion [3], [16], [17], and force balancing criterion [18], [19], [20], [21].

Vukobratovi et al. [14], [15] initially presented the ZMP criterion, which checks the stability by judging whether the ZMP lies within the support polygon. It has been widely used to evaluate legged robots' stability because of its clear physical meaning and easy-use advantage. Note that this criterion assumes that contact terrain is flat and leg-terrain friction is adequate. In addition, the ZMP criterion is a constraint to prevent the robot from tipping over. It only uses the support legs' rotating information, ignoring the swing legs' information. That said, it overlooks the robots' alternating support character.

The Poincare return map is a tool to transform the stability of a periodic orbit to a fixed point's stability [22], which can be used to judge periodically walking robots' stability. Grizzle et al. [16], [17] extended this criterion to a hybrid dynamics with impulse effects and applied it to judge the stability of a three degree-of-freedom (DoF) humanoid robot with leg-terrain impacts. Fu et al. [3] presented a section-map stability criterion to check the stability of nonperiodic walking robots.

The force balancing criterion relies on the force-balance principle to check whether the robot can tip over or slip. Hirukawa et al. [18] presented a novel stability criterion for legged robots, which checks if the sum of the gravity and inertia wrench lies within the region constructed by noncoplanar contact points. Although more universal than the ZMP criterion, it is still based on the sufficient friction assumption. Caron et al. [19] considered friction and linearized pendulum constraints and presented a more necessary and sufficient criterion to check the multicontact stability. Using the static force balancing model and optimization algorithm, Audren et al. [21] constructed a robust 3-D region that consists of all feasible center of mass (CoM) positions by suppressing bounded CoM accelerations. If the CoM position remains within this region, the robot is stable. These interesting stability criteria are the only necessary conditions for stable locomotion. The robot may still be stable due to its alternating support character when these conditions are violated.

In addition, the abovementioned criteria are limited to robots that are only driven by legs and cannot be applied to our robot.

Previously, considering the alternating support character, we proposed a stability criterion for our robot in [23], extending the concept of the capture point to a capture domain and determining stability based on the relationship between the capture and support domains. However, this criterion is only applicable to flat terrain walking. Besides, it requires both motion states and thrust values for stability evaluation, complicating the design of stability enhancement methods.

According to the difference in principle, the stability enhancement methods can be divided into three categories: robotic functional mechanism designing [24], [25], [26], [27], supernumerary tail adding [28], [29], [30], [31], [32], and inner motion planning and controlling [33], [34], [35], [36], [37], [38].

The first category focuses on modifying robots' geometrical and physical properties to buffer the leg-terrain impact or improve the self-stabilizing capability. Chellapurath et al. [24] disclosed how leg configuration, net weight, and the nature of the substrate affect the passive station-keeping performance of an underwater legged robot and found that the performance improves with an increase in net weight, a decrease in height, and an increase in width. In [25] and [26], two novel self-adaptive toe mechanisms are presented to possess cat claws' buffer and hold performance.

The second category involves using the drag or inertia force generated by the added tail to enhance stability. Zhang et al. [28] presented a new algorithm to alter the system center of gravity by adjusting the joint angles of a three DoF manipulator, enhancing the ability to prevent tip over. Inspired by basilisk lizards, Kim et al. [29] developed an interesting water-running robot that uses a two DoF tail to provide drag for enhancing the stability of yaw and pitch motion. In [30], an elegant optimal controller is proposed to steer the joints of the added tail to compensate for the neural delay dynamics for enhancing the human stance's stability. Using a deep reinforcement learning method, Huang et al. [31] created a policy to automatically adjust the behaviors of the tail to maintain a quadruped robot's stability.

The third category relies on the stability criterion or bioinspiration to determine control inputs and plan the robot's desired motion. In [33] and [34], the ZMP criterion is used to construct stability constraints integrated into the trajectory optimization framework to obtain optimal and stable motions for legged robots. Choi et al. [35] presented a new method to form the regions of attraction for underactuated legged robots, and it facilitates the design of the controller for stability enhancement. Using adaptive central pattern generators, Mehr et al. [36] presented a novel motion planner to adjust the upper body's position for ensuring a lower limb exoskeleton's postural stability. Imitating animals' natural gaits, Humphreys et al. [38] presented a gait transition method for locomotion proficiency enhancing.

Despite the abovementioned methods achieving impressive results in stability enhancement, they are mainly applicable to robots that allow the addition of tails or robots only driven by legs. They are challenging to apply to legged robots simultaneously driven by legs and thrusters.

B. Proposed Approach and Contribution

In this article, we establish a force balancing model and use the optimization method to find a feasible set of thrusts that allows for a certain degree of tip over and can also prevent our

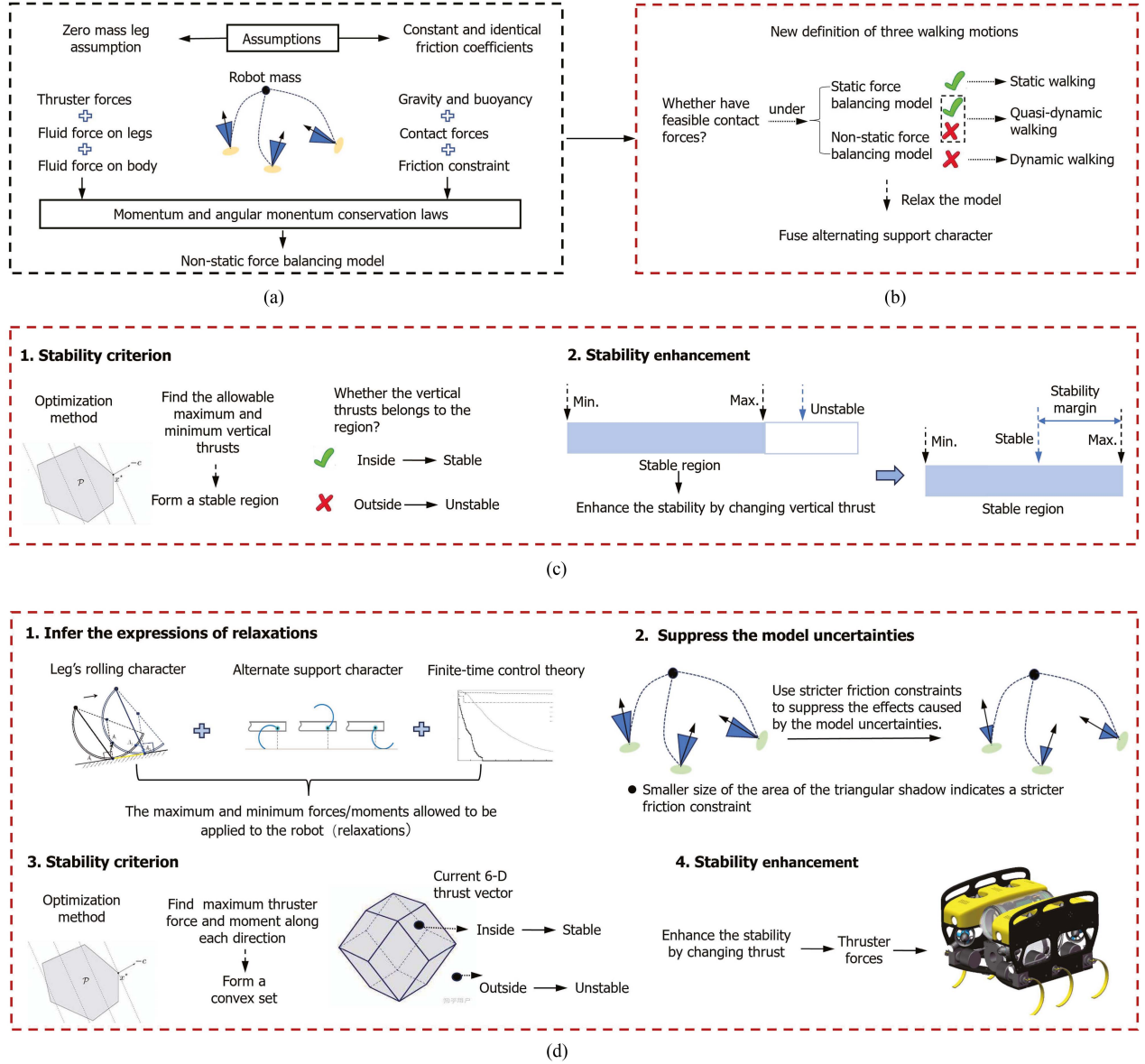


Fig. 2. Framework of the stability criterion and stability enhancement method. (a) We establish a force balancing model for our robot. (b) We present a new method to define three types of walking motions by relying on the static and nonstatic force balancing model and fuse the alternating support character by relaxing the model. (c) We use the optimization method to find the allowable maximum and minimum vertical thrusts and propose a stability criterion by checking whether the robot's vertical thrust belongs to the optimized stable region. Then, we present a stability enhancement method that changes the robot stability margin by adjusting the value of the vertical thrust. (d) We provide the expression of model relaxations using finite-time control theory to derive the maximum and minimum forces/moments. Then, we use stricter friction constraints to suppress the effects caused by the model uncertainties. Finally, we present a 6-D stability criterion and stability enhancement method. (a) Force balancing model. (b) Fuse alternating support character. (c) Vertical thrust based stability criterion and enhancement. (d) 6-D thrust stability criterion and enhancement.

robot from slipping. Using the allowable vertical thrust and 6-D thrust vector, we present two solutions to check and enhance the robot's stability. Fig. 2 shows the framework of the stability criterion and enhancement method. Table I shows the summary of related criteria and our criterion from the aspects of the used dynamic model, alternating support character, applied range for rough terrain, and whether to consider leg-terrain frictional constraint and joint torque limit. Apart from the Poincare return map criterion, the other related criteria do not fuse the alternating support character. In addition, except for the force balancing criterion, the other related criteria are unsuitable for judging the stability of rough terrain climbing and do not consider the

leg-terrain frictional constraint. Finally, all related criteria do not consider the joint's torque limit. In summary, our criterion can compensate for the deficiencies of the related criteria above. In addition, once the ZMP position is not in the support region of a legged robot, the ZMP criterion indicates that the robot will tip over, part of the legs will lift off the ground and then their contact forces will become zero. The force balancing criterion judges the robot's stability by checking feasible contact forces' existence. Both criteria use the force-balance principle to check whether existing the feasible contact forces to judge the stability. Poincare return map and capture theory based criteria use the swing and support legs' rotational states, which judge stability

TABLE I
COMPARISON OF RELATED CRITERIA AND OUR CRITERION

Criteria	References	Used model	Fuse alternating support character	Suitable for rough terrain	Consider leg-terrain frictional constraint	Consider joint torque limit
ZMP criterion	[14], [15]	Inverted pendulum	×	×	×	×
Poincare return map criterion	[3], [16], [17]	Inverted pendulum	✓	×	×	×
Force balancing criterion	[18]–[21]	Force balancing	×	✓	✓	×
Capture theory based criterion	[23]	Inverted pendulum	✓	×	×	×
Ours		Force balancing	✓	✓	✓	✓

by checking whether the robot's motion states belong to the prescribed set.

This article's difficulty lies in how to propose the method of defining static, quasi-dynamic, and dynamic walking motions for thruster-assisted legged robots. Moreover, how to judge the robot's stability by using the defined motions and considering the alternating support character is another difficulty.

The main contributions can be specified as follows.

- 1) We establish a force balancing model of our robot by considering the fluid forces, hip joint's driving torque limit, friction limit, and for rough terrain climbing.
- 2) We define three motions under the force balancing model, e.g., static, quasi-dynamic, and dynamic walking, and reveal the connection associated with the alternating support character.
- 3) We present a stability criterion by using an optimization algorithm to find a feasible set of thrusts, which has a good necessity due to considering the alternating support character.
- 4) We propose a stability enhancement method with clear physical meanings because the desired stability margin can be achieved only by adjusting the thruster force according to the feasible set.

The rest of this article is organized as follows. Section II establishes a force balancing model for our robot. Section III defines three motions under the established model and presents a model relaxation method to fuse the alternating support character. Section IV provides a vertical thrust-based stability criterion and stability enhancement method. Section V presents a 6-D thrust vector based stability criterion and stability enhancement method. Sections VI and VII present simulation and experimental results, respectively. Finally, Section VIII concludes this article.

II. FORCE BALANCING MODEL

A. Framework of the Force Balancing Model

Assumption 1: The legs have zero mass, and the robot's CoM is at the plane's center point, consisting of six hip joints.

Assumption 2: The shape of the leg is a semicircle, and its deformation is ignored.

Assumption 3: Friction coefficients between legs and terrain are constant and identical.

Assumption 4: The drive motors of the six legs are identical.

Since the total mass of the six legs accounts for about 3% of the robot's mass, its influence on the robot's dynamics is negligible. In addition, the C-shaped legs are made of hard steel. Their shape deformations are extremely tiny while impacting terrains. Thus, we ignore the legs' deformations. The unmodeled

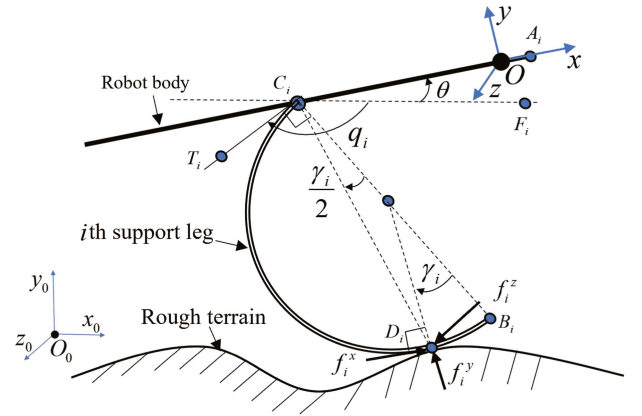


Fig. 3. Diagram of i th supporting leg that touches rough terrain.

dynamics caused by ignoring the legs' masses can make the established force balancing model unable to accurately reflect the robot's motion. Therefore, it cannot yield the accurate maximum and minimum thruster forces, and the accuracy of stability judgement can be reduced. However, the unmodeled dynamics caused by ignoring legs' masses are relatively small, and their adverse effects can be ignored.

According to whether touching the rough terrain, the six legs can be divided into two categories: supporting legs and swing legs. Fig. 3 shows the diagram of i th supporting leg that touches rough terrain. $O_0x_0y_0z_0$ and $Oxyz$ are earth-fixed and body-fixed frames. A_i and B_i represent the front and end points of the robot body and i th supporting leg, C_i and D_i represent the hip joint point and touchdown point, r is the radius of the leg, q_i represents the rotation angle, where $q_i = \angle F_i C_i T_i$ and the segment $C_i T_i$ is perpendicular to $C_i B_i$. $\gamma_i = 2\angle B_i C_i D_i$ is the support angle of the leg. θ is the robot's pitch angle. f_i^x , f_i^y , and f_i^z are forward, vertical, and lateral touching forces.

Using momentum and angular momentum conservation laws, we obtain the framework as

$$\begin{cases} \dot{Q} + w \times Q = F_B + F_G + F_b^f + F_b^f + F_T + A_f f \\ \dot{L} + w \times L + v \times Q = M_{BG} + M_b^f + M_l^f + M_T + A_m f \end{cases} \quad (1)$$

$$\|B_t f_i\| \leq u^\top(\mu) f_i \leq \Gamma_i, \quad i = 1, \dots, N_s \quad (2)$$

where $Q = m(v + w \times r_c) \in \mathbb{R}^3$ is the momentum of the robot; $L = J_0 w + m r_c \times v \in \mathbb{R}^3$ is the angular momentum of the robot; $v \in \mathbb{R}^3$ and $w \in \mathbb{R}^3$ are velocity and angular velocity of the robot buoyant center; $m \in \mathbb{R}$ is the mass of the robot, $J_0 \in \mathbb{R}^{3 \times 3}$ is the inertial matrix of the robot; $r_c \in \mathbb{R}^3$ is the

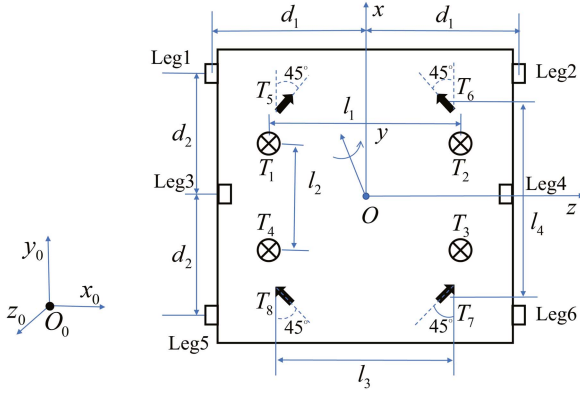


Fig. 4. Placement of thrusters and legs [5].

position vector from the buoyant center to CoM; $F_B \in \mathbb{R}^3$ and $F_G \in \mathbb{R}^3$ are the buoyancy and gravity vectors; $M_{BG} \in \mathbb{R}^3$ is the moment caused by buoyancy and gravity; $F_b^f \in \mathbb{R}^3$ and $M_b^f \in \mathbb{R}^3$ are the fluid force and fluid moment applied to robot body; $F_l^f \in \mathbb{R}^3$ and $M_l^f \in \mathbb{R}^3$ are the fluid force and fluid moment applied to C-shaped legs; $F_T \in \mathbb{R}^3$ and $M_T \in \mathbb{R}^3$ are the force and moment caused by eight thruster forces; $f = [f_1^\top, \dots, f_i^\top, \dots, f_{N_s}^\top]^\top \in \mathbb{R}^{3N_s}$ is the set of contact forces of N_s supporting legs; $f_i = [f_i^x, f_i^y, f_i^z]^\top \in \mathbb{R}^3$ is the contact force of i th supporting leg; $A_f \in \mathbb{R}^{3 \times 3N_s}$ and $A_m \in \mathbb{R}^{3 \times 3N_s}$ are the contact matrices that sum all contact forces and contact moments; $B_t = \text{diag}(1, 0, 1) \in \mathbb{R}^{3 \times 3}$ is the friction cone matrix, $\text{diag}(\cdot)$ is a function to describe the diagonal matrix; $u = [0, \mu, 0]^\top \in \mathbb{R}^3$ is the matrix representing the contact normals and the friction coefficient μ ; $\|\cdot\|$ is the Euclidian norm; Γ_i is a positive constant, which is related to the torque limit of i th supporting leg's hip joint.

Next, we need to further model, as follows:

- 1) the forces, such as F_T , M_T , F_B , F_G , and M_{BG} , caused by thrusters, buoyancy, and gravity;
- 2) the fluid forces, such as F_b^f , M_b^f , F_l^f , and M_l^f , related to the robot motion states;
- 3) the contact matrices A_f and A_m that are associated with robot's touching state and terrain's outline;
- 4) the positive constant Γ_i related to i th hip joint's torque limit.

Remark 1: The established model in (1) shows that only one touching point is required for each support leg. To ensure this, the curvature radius of the terrain near the touching point should be larger than the leg's radius. In addition, the terrain cannot be sticky because the established model is based on nonnegative vertical contact forces. Once it is sticky, the vertical contact force f_i^y can be less than zero, and the friction constraint in (2) cannot be satisfied.

B. Thruster Forces, Gravity, and Buoyancy

According to the thruster placement in Fig. 4, we obtain the force F_T and moment M_T caused by eight thrusters as

$$[F_T^\top, M_T^\top]^\top = C_t \cdot T \quad (3)$$

where $F_T = [F_{Tx}, F_{Ty}, F_{Tz}]^\top$, $M_T = [M_{Tx}, M_{Ty}, M_{Tz}]^\top$, $T = [T_1, \dots, T_8]^\top \in \mathbb{R}^8$ is the thruster force vector, $C_t \in \mathbb{R}^{6 \times 8}$ is the force allocation matrix [4] and it is defined as

$$\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{l_1}{2} & -\frac{l_1}{2} & -\frac{l_1}{2} & \frac{l_1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}l_{34}}{4} & \frac{\sqrt{2}l_{34}}{4} & \frac{\sqrt{2}l_{34}}{4} & -\frac{\sqrt{2}l_{34}}{4} \\ \frac{l_2}{2} & \frac{l_2}{2} & -\frac{l_2}{2} & -\frac{l_2}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$$

where l_1 , l_2 , and l_3 are three geometrical distances between thrusters, which are shown in Fig. 4, and $l_{34} = l_3 + l_4$.

The robot body's gravity and buoyancy in the body-fixed frame can be expressed as

$$F_B = \begin{bmatrix} B \sin \theta \\ B \cos \theta \cos \phi \\ -B \cos \theta \sin \phi \end{bmatrix}, F_G = \begin{bmatrix} -G \sin \theta \\ -G \cos \theta \cos \phi \\ G \cos \theta \sin \phi \end{bmatrix} \quad (4)$$

where ϕ is the robot's roll angle.

We can deduce the moment vector caused by buoyancy and gravity in the body-fixed frame as

$$M_{BG} = r_c \times F_G \quad (5)$$

C. Fluid Forces Applied to Robot Body and Legs

The fluid forces applied to the robot body are described as

$$\begin{cases} F_b^f = -\text{diag}(\lambda_1, \lambda_2, \lambda_3)\dot{v} - \text{diag}(c_x, c_y, c_z)v \\ M_b^f = -\text{diag}(\lambda_4, \lambda_5, \lambda_6)\dot{w} - \text{diag}(m_x, m_y, m_z)w \end{cases} \quad (6)$$

where λ_1 , λ_2 , and λ_3 are added masses along the robot body's longitudinal, vertical, and lateral directions, λ_4 , λ_5 , and λ_6 are added moments of inertia along the three directions. c_x , c_y , c_z and m_x , m_y , m_z are two groups of hydrodynamic parameters linearly related to water drags and moments of resistance.

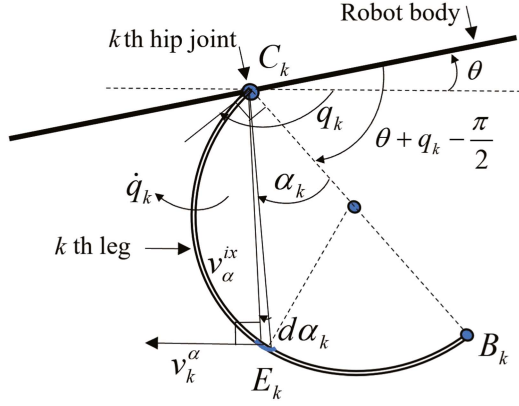
Remark 2: Because our robot's cruising speed is lower than that of torpedo-like underwater robots, we believe that the first-order fluid force accounts for most of the hydrodynamics and then ignores the second or higher order fluid force. In addition, these hydrodynamic parameters such as $\lambda_1, \dots, \lambda_6$, c_x , c_y , c_z , m_x , m_y , and m_z are identified from experimental data by multiple pool cruising tests. In addition, the value of the angular acceleration $\dot{w}$ cannot be measured directly by sensors. Here, we use the value of the robot's current and previous angular velocities and the difference method to obtain the value of $\dot{w}$.

Before deducing the fluid forces applied to the legs, we need to know the velocities of the legs. In addition, we need to lump all fluid forces into the buoyancy centre. Here, we split the deducing process into three steps: deduce the legs' velocities, deduce the fluid forces, and lump the forces into the buoyancy centre.

Step 1: Deduce the legs' velocities. Based on the placement of the hip joints in Fig. 4 and the velocity v and angular velocity w in the buoyant centre, we can deduce the velocity of k th hip joint as

$$v_k^h = v + w \times r_k^h, \quad k = 1, \dots, 6 \quad (7)$$

where $r_1^h = [d_2, 0, -d_1]^\top$, $r_2^h = [d_2, 0, d_1]^\top$, $r_3^h = [0, 0, -d_1]^\top$, $r_4^h = [0, 0, d_1]^\top$, $r_5^h = [-d_2, 0, -d_1]^\top$, and $r_6^h = [-d_2, 0, d_1]^\top$,

Fig. 5. Velocity of one short arc that belongs to k th leg.

d_1 and d_2 are two kinds of distances between hip joints and are shown in Fig. 4.

As shown in Fig. 5, we provide the diagram of a short arc and its velocity v_k^α in k th leg. Based on the geometrical relationship, we obtain the velocity v_k^α as

$$v_k^\alpha = v_k^h + [0, 0, -\dot{q}_k]^\top \times r_k^\alpha \quad (8)$$

where $r_k^\alpha = [2r \cos \alpha_k \cdot \cos(\theta + q_k + \alpha_k - \frac{\pi}{2}), -2r \cos \alpha_k \cdot \sin(\theta + q_k + \alpha_k - \frac{\pi}{2}), 0]^\top$, $\alpha_k = \angle B_k C_k E_k$.

Step 2: Deduce fluid forces. The fluid force applied to the short arc can be described as

$$dF_k^\alpha = \text{diag}(-\mu_l, -\mu_l, 0) v_k^\alpha \cdot d\alpha_k \quad (9)$$

where μ_l is the hydrodynamic coefficient related to the water density and C-shaped leg's width. Note that we ignore the fluid force along the leg's lateral direction because the C-shaped leg's width is much larger than its thickness.

The moment on the k th hip joint caused by dF_k^α can be deduced as

$$dM_k^\alpha = r_k^\alpha \times dF_k^\alpha. \quad (10)$$

Integrating both sides of (9) and (10), we can deduce the fluid force and moment applied to the i th hip joint as

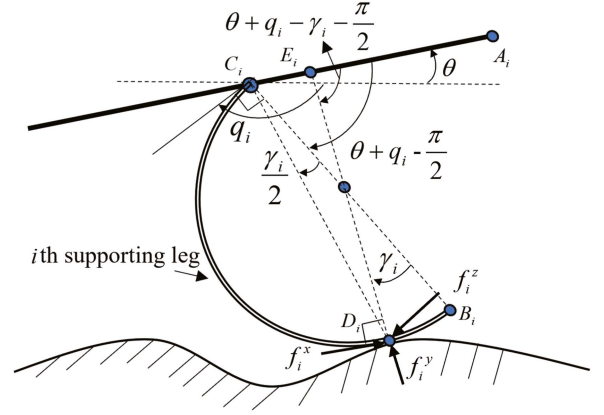
$$\begin{cases} F_k^\alpha = \int_0^{\frac{\pi}{2}} \text{diag}(-\mu_l, -\mu_l, 0) v_k^\alpha \cdot d\alpha_k \\ M_k^\alpha = \int_0^{\frac{\pi}{2}} r_k^\alpha \times \text{diag}(-\mu_l, -\mu_l, 0) v_k^\alpha \cdot d\alpha_k. \end{cases} \quad (11)$$

Step 3: Lump the forces into the buoyancy center. Lumping the forces on six hip joints, we can deduce the force and moment applied to the buoyancy center as

$$F_l^f = \sum_{i=1}^6 F_k^\alpha, \quad M_l^f = \sum_{i=1}^6 (M_k^\alpha + r_k^h \times F_k^\alpha). \quad (12)$$

D. Contact Matrices and Torque Limit

Fig. 6 shows the contact force $f_i = [f_i^x, f_i^y, f_i^z]^\top$ of i th supporting leg, where $i = 1, 2, \dots, N_s$, f_i^y and f_i^x are parallel and perpendicular to the line segment $D_i E_i$, respectively, f_i^z is parallel to the axis Oz of the body-fixed frame, C_i and D_i are points of i th hip joint and touching with the rough terrain, and E_i is the point that D_i projects to the robot body along the direction of f_i^y .

Fig. 6. Contact force of the k th supporting leg.

Since $\angle A_i C_i B_i = \theta + q_i - \frac{\pi}{2}$, we can derive the position vector from C_i to D_i as $r_i^t = [l_i^{cd} \cos \angle A_i C_i D_i, -l_i^{cd} \sin \angle A_i C_i D_i, 0]^\top$, where $l_i^{cd} = 2r \cos(\frac{\gamma_i}{2})$, $\angle A_i C_i D_i = \angle A_i C_i B_i + \frac{\gamma_i}{2} = \theta + q_i - \frac{\pi}{2} + \frac{\gamma_i}{2}$.

Then, we derive the contact matrices A_f and A_m as

$$\begin{cases} A_f = \text{diag}(J_1, J_2, \dots, J_{N_s}) \\ A_m = \text{diag}(\hat{r}_1^t J_1, \hat{r}_2^t J_2, \dots, \hat{r}_{N_s}^t J_{N_s}) \end{cases} \quad (13)$$

where the hat map $(\cdot)$ is used to turn a vector into a skew-symmetric matrix, J_i denotes a rotation matrix to project the f_i into the body-fixed frame, which is expressed

$$J_i = \begin{bmatrix} \cos(\angle A_i E_i D_i - \frac{\pi}{2}) & \sin(\angle A_i E_i D_i - \frac{\pi}{2}) & 0 \\ -\sin(\angle A_i E_i D_i - \frac{\pi}{2}) & \cos(\angle A_i E_i D_i - \frac{\pi}{2}) & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

where $\angle A_i E_i D_i = \angle A_i C_i B_i + \gamma_i = \theta + q_i - \frac{\pi}{2} + \gamma_i$.

According to the geometrical relationship in Fig. 6, we can derive the needed torque of the driving motor in i th hip joint as

$$\tau_i = l_i^{cd} f_i^x \cos\left(\frac{\gamma_i}{2}\right) + l_i^{cd} f_i^y \sin\left(\frac{\gamma_i}{2}\right). \quad (14)$$

Equation (14) can be rewritten as

$$\tau_i = l_i^{cd} \sqrt{f_i^x{}^2 + f_i^y{}^2} \sin\left(\frac{\gamma_i}{2} + \alpha_i\right) \quad (15)$$

where $\alpha_i = \arctan(f_i^x / f_i^y)$.

Since $\sin(\frac{\gamma_i}{2} + \alpha_i) \leq 1$ and $f_i^x \leq \mu |f_i^y|$, we have

$$\tau_i \leq l_i^{cd} \sqrt{1 + \mu^2} \cdot u^\top(\mu) f_i \leq \tau_{\max} \quad (16)$$

where $\tau_{\max}$ is the maximum torque that the driving motor in i th hip joint can be provided.

Based on (16), we have $u^\top(\mu) f_i \leq \frac{\tau_{\max}}{l_i^{cd} \sqrt{1 + \mu^2}}$. Thus, we can obtain i th supporting leg's Γ_i as

$$\Gamma_i = \frac{\tau_{\max}}{l_i^{cd} \sqrt{1 + \mu^2}}. \quad (17)$$

III. WALKING MOTION DEFINITION

We introduce the existing definition of the humanoid robot's walking motions, which uses the relation between ZMP, CoM, and the support area to judge its stability. Inspired by this, we propose a new method to define the walking motions for

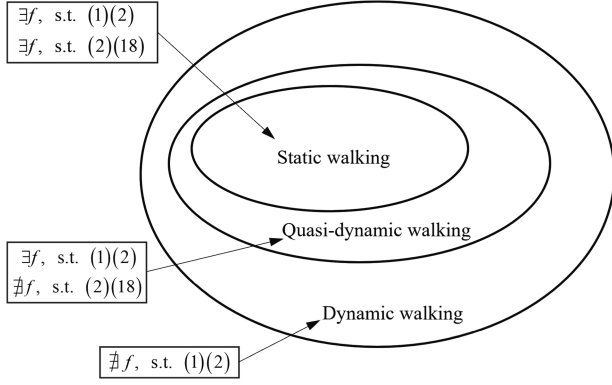


Fig. 7. Diagram of static, quasi-dynamic, and dynamic walking motions.

thruster-assisted legged robots using the established force balancing model to check whether there are feasible contact forces. Finally, we fuse the alternating support character by relaxing the established model as inequality constraints.

According to the relationship between the CoM, ZMP and the support convex polygon, Chen et al. [39] define three types of humanoid robot walking motions. Concretely, when the CoM and ZMP are located in the support convex polygon area, the robot belongs to *static walking motion*. When the CoM is located in the region of the supporting convex polygon, and the ZMP is outside the polygon, the robot is in *quasi-dynamic walking motion*. The robot is in *dynamic walking motion* when the ZMP is located outside the support convex polygon.

Before defining the motions, we remove the inertia and fluid forces from (1) and obtain the static force balancing model as

$$\begin{cases} F_B + F_G + A_f f + F_T = 0 \\ M_{BG} + A_m f + M_T = 0. \end{cases} \quad (18)$$

According to whether there are feasible contact forces f to make (1), (2), and (18) satisfy, we can sort the motions into three categories: static, quasi-dynamic, and dynamic walking motions, which are shown in Fig. 7.

Definition 1 (Static Walking): If $\exists f$, satisfying (1), (2), and also $\exists f$, satisfying (2), (18), the robot is in static walking motion.

Definition 2 (Quasi-dynamic Walking): If $\exists f$, satisfying (1) and (2), but $\nexists f$ satisfies (2) and (18), then the robot is in quasi-dynamic walking motion.

Definition 3 (Dynamic Walking): If $\nexists f$, satisfying (1) and (2), then the robot is in dynamic walking motion.

Remark 3: Compared to the definition of the three types of motions in [39], our definition has the advantage of being close to the actual situation and can be applied to more universal cases. The advantages lie in the following three aspects.

- 1) Unlike the definition in [39], in addition to being relevant to the gravity and inertia force, our definition is also relevant to the fluid and thruster forces. Therefore, our definition is more universal and can be used to define the motion of thruster-assisted robots and underwater-legged robots.
- 2) Different from the definition in [39], which assumes that leg-terrain friction and joints' driving torques are large enough, our definition relaxes these assumptions. That said, the walking motions sorted by our definition are closer to the actual situation.

- 3) Our definition sorts the walking motions by checking whether feasible leg-terrain contact forces exist. Therefore, it is convenient to sort the robot's motion of climbing rough terrain. In contrast, the definition in [39] needs to assume that the terrain is flat. In addition, the use of a contact force-based force balancing model makes it easy to fuse other kinds of external forces and thrusts.

We may be unable to find the feasible contact forces (solutions) by using optimization methods when the constraints in (1) and (2) are too strict. Many factors lead to the constraints' strictness, e.g., the number of support legs, the amount of inertial, fluid forces, and thrusts. Next, the existence of feasible solutions when the robot has 1, 2, and at least three support legs will be discussed, respectively.

- 1) When the robot has only one support leg, the number of unknown variables in (1) is 3, and there are three contact forces at the touchdown point. The number of the equality constraints in (1) is 6. Thus, (1) is overconstrained. That said, there is no feasible contact force. In such a case, the robot belongs to dynamic walking.
- 2) When the robot has two support legs, the number of unknown variables in (1) is 6, which is the number of contact forces at the two touchdown points. The number of inequality and equality constraints are 4 and 6, respectively. Thus, it is highly likely that a feasible solution for six variables cannot be found. In such a case, the robot is highly dynamic walking.
- 3) When the robot has at least N_s ($N_s \geq 3$) support legs, the number of unknown variables in (1) is $3N_s$. The number of equality constraints in (1) is 6, and the number of inequality constraints is $2N_s$. Thus, when the inertial forces and the thrusts are within a reasonable range, feasible solutions in $3N_s$ variables can be found, and the robot may belong to static or dynamic walking.

When it belongs to dynamic walking, and its swing leg does not switch to the support one, the legged robot must fall due to tip over. However, the legged robot relies on the legs' alternate support character to realize the walking. It is still possible to make the swing leg switch to the support leg timely by gait planning, and the robot remains stable.

If the robot's stability is judged only by the existence of feasible solutions, the necessity of the stability criterion will be reduced. Therefore, the equality constraints in (1) should be appropriately relaxed to improve the necessity of the stability criterion as follows.

$$\begin{cases} -\Omega_F \leq -\dot{Q} - w \times Q + (F_B + F_G + F_b^f + F_l^f \\ \quad + A_f q_f + A_f f + F_T) \leq \Omega_F \\ -\Omega_M \leq -\dot{L} - w \times L - v \times Q + (M_{BG} + M_b^f \\ \quad + M_l^f + A_m q_f + A_m f + M_T) \leq \Omega_M \end{cases} \quad (19)$$

where $\Omega_F = [\Omega_F^1, \Omega_F^2, \Omega_F^3]^\top \in \mathbb{R}^3$ and $\Omega_M = [\Omega_M^1, \Omega_M^2, \Omega_M^3]^\top \in \mathbb{R}^3$ are relaxations, $\Omega_F^2 = \Omega_M^2 = 0$, $\Omega_F^1, \Omega_F^3, \Omega_M^1$, and Ω_M^3 are positive. Note that the values of relaxations reflect the degree of losing the model equation in (1) into an inequality in (19), and the relaxations contribute to fusing the legged robot's alternating support character.

The magnitude of the relaxation is related to the leg's alternate touching/swing smoothness and the robot's mass. When the alternation is smoother, the relaxation should be smaller. In addition, if the vertical force of the robot body is relaxed, the supporting leg may leave the ground, and the walking property

of the legged robot may be lost. Thus, the relaxation amount Ω_F^2 is set to zero to maintain the vertical balance. Finally, the magnitude of Ω_M^2 represents the degree of losing the yawing motion, which is unrelated to the alternating support character. Then, we set $\Omega_M^2 = 0$.

IV. VERTICAL THRUST-BASED STABILITY CRITERION AND ENHANCEMENT

We present a criterion to judge the robot's stability by checking whether the vertical thrust belongs to the range of allowable vertical forces. On this basis, we propose a method to enhance the stability by changing the vertical thrust's value.

When the vertical thrust is too large, the robot may fall due to the support leg's torque limit. When the vertical thrust is too small, the forward and lateral contact forces may not be sufficient to suppress the robot's inertial and fluid forces, resulting in the support leg slipping. Setting the appropriate vertical thrusts is necessary so the robot will not fall or slip. We need to find the maximum/minimum vertical thrusts that can be applied to the robot body without losing stability.

Specifically, we transform the problem of exploring the maximum/minimum vertical thrusts, e.g., $F_{Ty}^{\max}$ and $F_{Ty}^{\min}$, into a second-order cone programming problem. Then, we use the sequential least squares programming algorithm to find the optimal solutions

$$\begin{cases} \max & F_{Ty} \text{ or } -F_{Ty} \\ \text{s.t.} & (2) \text{ and } (19). \end{cases} \quad (20)$$

Proposition 1: When the vertical thrust is greater than $F_{Ty}^{\min}$ or less than $F_{Ty}^{\max}$, i.e., $F_{Ty} \in [F_{Ty}^{\min}, F_{Ty}^{\max}]$, the robot is stable, and its stability margin is $\Psi = \min(F_{Ty} - F_{Ty}^{\min}, F_{Ty}^{\max} - F_{Ty})$. Otherwise, it is unstable.

When the robot climbs rough terrains, it will inevitably suffer from various disturbances caused by ocean currents and cables. The disturbances may make the robot lose its balance. Unlike the conventional ground legged robots, our robot is actuated by thrusters and legs simultaneously, where thruster forces can alter the robot's dynamics and break or maintain the robot's balance. Thus, we present a solution to find the set of feasible thruster forces to keep the robot's dynamic equilibrium and prevent the legs' slippage.

The vertical thrust can be set as the average of the obtained maximum/minimum vertical thrust to make the robot in the most stable state

$$F_{Ty} = \frac{F_{Ty}^{\min} + F_{Ty}^{\max}}{2}. \quad (21)$$

Note that the stability margin of the robot with this vertical thrust is maximum, which can be expressed as $\Psi_{\max} = \frac{F_{Ty}^{\max} - F_{Ty}^{\min}}{2}$. In fact, we do not request the robot walk with maximum stability margin because stability and flexibility conflict. Once the robot is in the most stable state, it may be unable to maneuver flexibly. Then, we need to balance the two conflicting requests by setting the stability margin.

Specifically, once the robot is requested to walk with the desired stability margin $\Psi_d \in [0, \Psi_{\max}]$, we can set the vertical thrust as

$$F_{Ty} = F_{Ty}^{\max} - \Psi_d \text{ or } \Psi_d - F_{Ty}^{\min}. \quad (22)$$

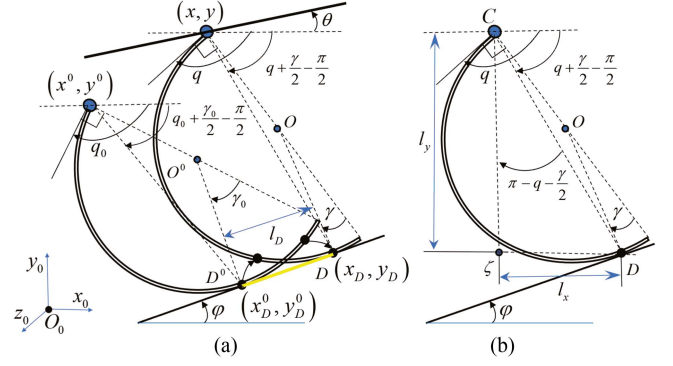


Fig. 8. Motion of rolling on the terrain.

Remark 4: We use the optimization method to strip the feasible thrusts from multiple variables. Thus, the robot's stability can be judged by checking whether the vertical thrust belongs to the region that consists of maximum and minimum thruster forces. Unlike the criteria that use multiclass states with different physical meanings, our criterion only relies on one physical quantity (vertical thrust). Thus, it has precise physical meanings and is also convenient for designing the stability enhancement method.

Remark 5: Although it possesses the above advantages, the criterion in this section only uses vertical thrust. Once the robot cannot maintain the prescriptive balance by adjusting the vertical thrust, the criterion regards that the robot is unstable. However, the robot may still be stable when using the 6-D thrust. That said, the criterion is slightly conservative. Thus, we need to use the 6-D thrust to improve its necessity and the accuracy of judging the robot's stability. In addition, the relaxations in this section are constant and designed artificially. They should be time-varying. Thus, we need to derive the expressions of these relaxations.

V. 6-D THRUST-BASED STABILITY CRITERION AND ENHANCEMENT

This section presents two stability criteria and their corresponding stability enhancement method by using the robot's eight thrusters, which can generate 6-D thruster forces and moments. First, we provide the method to derive the expressions of relaxations. Second, we present a method to produce allowable thruster forces and moments. Then, we propose a stability criterion and a method to enhance stability. Finally, we propose a 6-D thrust based robust stability criterion.

A. Derivation of Relaxations

This section establishes a 3-D pendulum's dynamic model to represent the robot's longitudinal and lateral motions under external forces. In addition, we deduce the allowable forces and moments according to the robot's allowable swing. Finally, we deduce the relaxations' expressions.

As shown in Fig. 8, the pendulum comprises one virtual supporting leg and one robot body, whose endpoint is fixed on the buoyant centre. The legs' touch angle γ and rotation angle q are the average values of all supporting legs, such as $\gamma = \sum_{i=1}^{N_s} \gamma_i$ and $q = \sum_{i=1}^{N_s} q_i$. γ_0 and q_0 are initial touch and rotation angles. (x^0, y^0) and (x, y) are initial and current positions of the pendulum. We also assume that the segment of

D_0D is a straight line, and its slant angle is φ . In addition, F_x, F_y , and F_z are forward, vertical, and lateral external forces exerted on the body. M_x and M_z are roll and pitch external moments, respectively. These forces are in the direction of the body-fixed frame.

Actuated by these external forces, the 3-D pendulum's dynamic model can be established as follows:

$$\begin{cases} ml_y\ddot{x} + ml_x\ddot{y} = [F_x \cos \theta + F_y \cos(\theta + 0.5\pi)]l_y \\ \quad + [F_x \sin \theta + F_y \sin(\theta + 0.5\pi)]l_x - M_z \\ ml_y^2\ddot{\zeta} = F_z l_y + M_x \end{cases} \quad (23)$$

where ζ is the virtual supporting leg's dip angle, $l_x = 2r \cos(0.5\gamma) \sin(\pi - q - 0.5\gamma)$, and $l_y = 2r \cos(0.5\gamma) \cos(\pi - q - 0.5\gamma)$.

Remark 6: Because the allowable variation range of dip angle is slight and often less than 10° , the coupling between the longitudinal and lateral motions is small. Thus, we split the 3-D locomotion into longitudinal and lateral motions. Moreover, the accelerations $\ddot{x}$ and $\ddot{y}$ are coupled, and they are related to the leg's rotation angle q and the slant angle φ .

Fig. 8(a) shows that the leg rolls from the initial touch point D^0 to the current point D on the slope. Then, we obtain the position of the point D as

$$x = x_D^0 + l_D \cos \varphi - l_x, \quad y = y_D^0 + l_D \sin \varphi + l_y \quad (24)$$

where l_D is the length of the segment D_0D .

The local surface that the leg rolls during a complete cycle cannot change dramatically, and it is assumed as a slope. Since the segments O^0D^0 and OD are parallel to the segment D^0D , we have

$$q_0 - 0.5\pi + \gamma_0 = q - 0.5\pi + \gamma. \quad (25)$$

Thus, we obtain that the length of the segment D_0D as

$$l_D = r(\gamma_0 - \gamma) = r(q - q_0). \quad (26)$$

Taking the derivative of both sides of (24) and using the definitions of l_x and l_y , we have

$$\dot{x} = r \cos \varphi \dot{q} - r \cos q \dot{q}, \quad \dot{y} = r \sin \varphi \dot{q} + r \sin q \dot{q}. \quad (27)$$

Taking the derivative of (27), we have

$$\begin{cases} \ddot{x} = r \cos \varphi \ddot{q} - r \cos q \ddot{q} + r \sin q \dot{q}^2 \\ \ddot{y} = r \sin \varphi \ddot{q} + r \sin q \ddot{q} + r \cos q \dot{q}^2. \end{cases} \quad (28)$$

Substituting (28) into (23) and based on the fact of $q + \varphi + \gamma = \pi$, we can rewrite the 3-D pendulum model as

$$2mr^2(\cos \gamma + 1)\ddot{q} + mr^2 \sin \gamma \dot{q}^2 = u_1, \quad ml_y^2\ddot{\zeta} = u_2 \quad (29)$$

where $u_1 = (F_x \cos \theta - F_y \sin \theta)l_y + (F_x \sin \theta + F_y \cos \theta)l_x - M_z$ and $u_2 = F_z l_y + M_x$.

Next, we will use the 3-D pendulum model, gait planning results, and the allowable lateral motion states to deduce the allowable forces and moments.

The gait planner provides all legs' rotation angles related to time. Then, we define that the virtual leg needs T_f seconds to make the q and $\dot{q}$ converge to q_d and $\dot{q}_d$, where T_f also means the time spent making the swing legs touching the terrain, and q_d is the supporting leg's rotation angle when it begins to leave the terrain and to turn into swing leg. In addition, the allowable lateral motion states are ζ_d and $\dot{\zeta}_d$, which are maximal dip angle and angular velocity at the moment of $t_0 + T_f$, where t_0 is the current moment.

In a word, we need to find the external forces F_x, F_y, F_z , and moments M_x, M_z to make the following conditions $\lim_{t \rightarrow t_0 + T_f} q(t) = q_d$, $\lim_{t \rightarrow t_0 + T_f} \dot{q}(t) = \dot{q}_d$, $\lim_{t \rightarrow t_0 + T_f} \zeta(t) = \zeta_d$, and $\lim_{t \rightarrow t_0 + T_f} \dot{\zeta}(t) = \dot{\zeta}_d$ hold, where q_d is determined by the gait planner, $\dot{q}_d$ is equal to $\Theta_{\min}$ or $\Theta_{\max}$ ($\Theta_{\min} < \Theta_{\max}$), ζ_d is equal to $\Upsilon_{\min}$ or $\Upsilon_{\max}$ ($\Upsilon_{\min} < \Upsilon_{\max}$), $\Theta_{\min}, \Theta_{\max}, \Upsilon_{\min}$, and $\Upsilon_{\max}$ are constants.

Thanks to the finite-time control theory in [40], we can find the external moments to satisfy the following conditions:

$$\begin{cases} u_1 = -\frac{ac_1 V_1^{\chi_1}}{\dot{q} - \dot{q}_d} + b\dot{q}^2 - a(q - q_d - \ddot{q}_d) \\ u_2 = -\frac{c_2 ml_y^2 V_2^{\chi_2}}{\dot{\zeta} - \dot{\zeta}_d} - ml_y^2(\dot{\zeta} - \dot{\zeta}_d - \ddot{\zeta}_d) \end{cases} \quad (30)$$

where χ_1 and χ_2 are positive constants, $0 < \chi_1 \leq \frac{1}{2}$ and $0 < \chi_2 \leq \frac{1}{2}$, $c_1 = \frac{V_1^{1-\chi_1}(t=t_0)}{T_f(1-\chi_1)}$, $V_1 = \frac{(q-q_d)^2 + (\dot{q}-\dot{q}_d)^2}{2}$, $a = 2mr^2(\cos \gamma + 1)$, $b = mr^2 \sin \gamma$, $c_2 = \frac{V_2^{1-\chi_2}(t=t_0)}{T_f(1-\chi_2)}$, $V_2 = \frac{(\zeta-\zeta_d)^2 + (\dot{\zeta}-\dot{\zeta}_d)^2}{2}$, $\ddot{q}_d$ and $\ddot{\zeta}_d$ are constants, u_1 and u_2 aim to make the equations such as $\dot{V}_1 = -c_1 V_1^{\chi_1}$ and $\dot{V}_2 = -c_2 V_2^{\chi_2}$ hold.

Because $\dot{V}_1 = (\dot{q} - \dot{q}_d)(q - q_d + \dot{q} - \dot{q}_d)$ and $\dot{V}_2 = (\dot{\zeta} - \dot{\zeta}_d)(\zeta - \zeta_d + \dot{\zeta} - \dot{\zeta}_d)$, $\dot{q} - \dot{q}_d$ and $\dot{\zeta} - \dot{\zeta}_d$ will not go to be zero until the V_1 and V_2 are going to zeros. Thus, $\dot{q} - \dot{q}_d$ and $\dot{\zeta} - \dot{\zeta}_d$ will not be zeros when $t \in (t_0, t_0 + T_f)$.

Lemma 1: The functions of u_1 and u_2 in (30) are monotonically decreasing in the variables of $\dot{q}_d$ and $\dot{\zeta}_d$, respectively. That said, $\frac{\partial u_1}{\partial \dot{q}_d} < 0$ and $\frac{\partial u_2}{\partial \dot{\zeta}_d} < 0$.

The Lemma's proof process can be seen in Appendix A.

Substituting the expressions of l_x and l_y into the expression of the u_1 behind (29), we can rewrite u_1 as

$$u_1 = 2r \cos(0.5\gamma) \cos(q + 0.5\gamma + \theta - \pi)F_x - 2r \cos(0.5\gamma) \sin(q + 0.5\gamma + \theta - \pi)F_y - M_z. \quad (31)$$

The angle $q + 0.5\gamma + \theta$ would be roughly around π because the robot body is substantially parallel to the segment of D^0D . Thus, we ignore the rolling dynamics caused by F_y . In addition, the u_1 is equally allocated to F_x and M_z , and the u_2 is equally assigned to F_z and M_x . In such cases, the external forces and moments are formulated as

$$\begin{cases} F_x = \frac{u_1}{4r \cos(0.5\gamma) \cos(q+0.5\gamma+\theta-\pi)} \\ F_y = 0, \quad F_z = \frac{u_2}{2l_y}, \quad M_z = -\frac{u_1}{2}, \quad M_x = \frac{u_2}{2}. \end{cases} \quad (32)$$

Since $\dot{q}_d \in [\Theta_{\min}, \Theta_{\max}]$ and $\dot{\zeta}_d \in [\Upsilon_{\min}, \Upsilon_{\max}]$, and using the conclusion in Lemma 1, we can derive that the maximum and minimum of F_x, F_z, M_x , and M_z in (32) as $F_x^{\max} = \max(F_x(\dot{q}_d = \Theta_{\min}), F_x(\dot{q}_d = \Theta_{\max}))$, $F_x^{\min} = \min(F_x(\dot{q}_d = \Theta_{\min}), F_x(\dot{q}_d = \Theta_{\max}))$, $F_z^{\max} = F_z(\dot{q}_d = \Theta_{\min})$, $F_z^{\min} = F_z(\dot{q}_d = \Theta_{\max})$, $M_x^{\min} = M_x(\dot{q}_d = \Theta_{\max})$, $M_x^{\max} = M_x(\dot{q}_d = \Theta_{\min})$, $M_z^{\min} = M_z(\dot{q}_d = \Theta_{\min})$, $M_z^{\max} = M_z(\dot{q}_d = \Theta_{\max})$, where $\min(\cdot, \cdot)$ and $\max(\cdot, \cdot)$ are functions to obtain the maximum and minimum values.

Then, using the maximum and minimum of F_x, F_z, M_x , and M_z , we can relax the robot dynamics equations in (1) and convert them to following inequations:

$$\begin{cases} F_{\text{low}} \leq -\dot{Q} - w \times Q + F_B + F_G + F_b^f + F_l^f \\ \quad + F_T + A_f f \leq F_{\text{up}} \\ M_{\text{low}} \leq -\dot{L} - w \times L - v \times Q + M_{BG} + M_b^f \\ \quad + M_l^f + M_T + A_m f \leq M_{\text{up}} \end{cases} \quad (33)$$

where $F_{\text{low}} = [F_x^{\min}, 0, F_z^{\min}]^\top$, $F_{\text{up}} = [F_x^{\max}, 0, F_z^{\max}]^\top$, $M_{\text{low}} = [M_x^{\min}, 0, M_z^{\min}]^\top$, $M_{\text{up}} = [M_x^{\max}, 0, M_z^{\max}]^\top$.

B. Stability Criterion and Enhancement

We choose a set of linearly independent unit vectors to represent the directions to find feasible maximum and minimum forces. Then, we regard the forces as vertices and construct a 6-D convex set to approximate the real feasible set for reducing the computational burden. Next, by using the established force balancing model, we formulate the problem of finding forces along the designed directions as an optimization problem

$$\begin{aligned} \max \quad & \vartheta_j^\top \xi \\ \text{s.t.} \quad & (33), (2) \end{aligned} \quad (34)$$

where $\vartheta_j \in \mathbb{R}^6$ ($j = 1, 2, \dots, N_\vartheta$) is the unit vector to denote the j th designed direction, $\xi = [F_T^\top, M_T^\top]^\top \in \mathbb{R}^6$.

After optimizing, we can obtain an extremal vertex called ξ_j^* along the search direction ϑ_j . Then, the 6-D feasible region can be approximated as a convex set $\mathcal{A}_n$, which is represented by the vertices representation method as

$$\mathcal{A}_n = \left\{ \sum_{j=1}^{N_\vartheta} \beta_j \xi_j^* \mid \beta_1 \geq 0, \dots, \beta_{N_\vartheta} \geq 0, \sum_{j=1}^{N_\vartheta} \beta_j = 1 \right\} \quad (35)$$

where β_j is the j th coefficient.

Proposition 2: The robot is stable if the vertex ξ , composed of the robot's thruster force and moment, belongs to $\mathcal{A}_n$. Otherwise, it is unstable.

Generally, the hyperplane representation method has the advantage of rapidly checking whether a vector is interior to a polytope by simply checking some given inequations and equations. However, it is difficult to represent a convex set if the number of vertices exceeds the vector's dimension number. Thus, we propose a stability checking method based on vertices representation.

Proposition 3: The vertex ξ belongs to $\mathcal{A}_n$ if there exist coefficients such as $\beta_1, \dots, \beta_{N_\vartheta}$ satisfy the following conditions:

- 1) $\xi = \sum_{j=1}^{N_\vartheta} \beta_j \xi_j^*$;
- 2) $\beta_1 \geq 0, \dots, \beta_{N_\vartheta} \geq 0$;
- 3) $\sum_{j=1}^{N_\vartheta} \beta_j = 1$.

Here, we use a linear programming algorithm to find the feasible coefficients as follows:

$$\begin{aligned} \text{find} \quad & \beta \\ \text{s.t.} \quad & \varrho \beta = \xi, \varpi^\top \beta = 1, \beta \succeq 0 \end{aligned} \quad (36)$$

where $\varpi = [1, 1, \dots, 1]^\top \in \mathbb{R}^{N_\vartheta}$, $\beta = [\beta_1, \dots, \beta_{N_\vartheta}]^\top \in \mathbb{R}^{N_\vartheta}$, $\varrho = [\xi_1^*, \dots, \xi_{N_\vartheta}^*] \in \mathbb{R}^{6 \times N_\vartheta}$.

Here, we propose two kinds of stability enhancement methods. The first method aims to make the robot walk with the maximum stability margin. The second one finds the stable thruster forces that closest approach to the previous thrusts.

The first method uses a quadratic programming algorithm to find the farthest thrust vector ξ' away from these vertices in the convex set

$$\min \sum_{i=1}^{N_\vartheta} \|\xi' - \xi_i^*\|_2^2$$

$$\text{s.t.} \quad \xi' = \sum_{j=1}^{N_\vartheta} \beta_j \xi_j^*, \varpi^\top \beta = 1, \beta \succeq 0 \quad (37)$$

where ξ' is the thrust vector to generate the maximum stability margin, the eight thrusters' forces can be calculated as $T = C_t^\dagger \xi'$, $C_t^\dagger$ is the pseudo-inverse matrix of C_t .

In the second method, we also use a quadratic programming algorithm to find the vector ξ' that not only inside the convex set but also is closest to previous thrust vector

$$\begin{aligned} \min \quad & \sum_{i=1}^{N_\vartheta} \|\xi' - \xi_{\text{pre}}\|_2^2 \\ \text{s.t.} \quad & \xi' = \sum_{j=1}^{N_\vartheta} \beta_j \xi_j^*, \varpi^\top \beta = 1, \beta \succeq 0 \end{aligned} \quad (38)$$

where ξ_{pre} is the thrust vector of the previous moment, which is called previous thrust vector, and the eight thrusters' forces can be described as $T = C_t^\dagger \xi'$.

Remark 7: When the robot is in the most stable situation, it may be unable to maneuver flexibly because stability and flexibility conflict. To balance the two conflicting factors, we present the second stability enhancement method, which decreases the variation of thrusts by obtaining the forces closest to the robot's previous thrusts. Note that the second method can steer the robot in a critical state once it is unstable in the previous moment.

C. Robust Stability Criterion and Enhancement

The force balancing model in (1) is not accurate enough because it ignores some dynamics, such as legs' inertia forces, high-order fluid forces, and external disturbances caused by ocean currents and the electrical cable.

The difficulty of providing an accurate enough robot model lies in the following three aspects.

- 1) The coupled dynamics between the legs and robot body increase the difficulty of modeling. They are challenging to decouple for stability analysis.
- 2) Unlike the torpedo-like robots with regular shapes and high locomotion velocity, our robot has an irregular open-frame structure and slow locomotion velocity. It is difficult to obtain accurate fluid forces for such a robot and scenario, especially the high-order fluid forces.
- 3) Ocean current varies in size from place to place, which is challenging to obtain a priori or measure online.

The unmodeled dynamics lead to an inaccurate convex set that may result in an inaccurate stability assessment. Thus, we formulate a stricter leg-terrain friction constraint to suppress uncertainties and present a robust stability criterion that will result in an accurate stability assessment.

The force balancing model that includes the unmodeled dynamics (uncertainties) $\Delta F \in \mathbb{R}^3$ and $\Delta M \in \mathbb{R}^3$ can be established as

$$\begin{cases} \dot{Q} + w \times Q = F_B + F_G + F_b^f + F_l^f + F_T + A_f f^j + \Delta F \\ \dot{L} + w \times L + v \times Q = M_{BG} + M_b^f + M_l^f + M_T + A_m f^j \\ + \Delta M, \quad \|B_t f_i^j\| \leq u^\top f_i^j \end{cases} \quad (39)$$

where ΔF and ΔM are bounded and satisfy that $\|\Delta F\| < r_f$, $\|\Delta M\| < r_m$, r_f and r_m are two known constants.

Theorem 1: If there exists a group of contact forces $f = [f_1^\top, \dots, f_{N_s}^\top]^\top \in \mathbb{R}^{3N_s}$ to make (40) true, we will find a group of contact forces f^j to make (39) true

$$\begin{cases} \dot{Q} + w \times Q = F_B + F_G + F_b^f + F_l^f + F_T + A_f f \\ \dot{L} + w \times L + v \times Q = M_{BG} + M_b^f + M_l^f + M_T + A_m f \\ \|B_t f_i\| \leq u^\top f_i + l_i - s_i \end{cases} \quad (40)$$

where l_i and s_i are two bounds, $s_i = (r_f + r_m)\bar{\sigma}(E_i^t A^\dagger)$, $\bar{\sigma}(\cdot)$ denotes the largest singular value of the matrix $(\cdot)$, $l_i = -(r_f + r_m) \|E_i^n A^\dagger\|$, i represents the sequence number for i th supporting leg, $A = [A_f^\top, A_m^\top]^\top \in \mathbb{R}^{6 \times 3N_s}$, $\dagger$ is the function to calculate the pseudoinverse matrix, $E_i^n = [0_{1 \times 3}, \dots, \underbrace{u^\top}_{i\text{th}}, \dots, 0_{1 \times 3}] \in \mathbb{R}^{1 \times 3N_s}$, $E_i^t = \text{diag}(0_{3 \times 3}, \dots, \underbrace{B_t}_{i\text{th}}, \dots, 0_{3 \times 3}) \in \mathbb{R}^{3 \times 3N_s}$.

That said, we can transfer the uncertainties on the robot body to the legs' contacting points and then use stricter friction constraints to suppress the effects caused by the uncertainties. This theorem's proof process is provided in Appendix B.

Inequation (40) is based on the scaling operations in (47) and (48) that aim to find their extreme values. Referring to [21], we state that these operations need to satisfy the following conditions.

- 1) Δ is perpendicular to the vectors of $E_1^n A^\dagger, \dots, E_{N_s}^n A^\dagger$.
- 2) Δ is collinear to the max singular vectors of $E_1^t A^\dagger, \dots, E_{N_s}^t A^\dagger$.

We cannot get the extreme values for each leg if these conditions cannot be satisfied simultaneously. Then, we use larger values to replace the effect caused by the bounded uncertainties ΔF_j and ΔM_j . It can reduce the tightness and increase the conservativeness of the formulated constraint, leading to a more conservative stability region.

Next, we will propose a method that finds an integrated realizable extreme value and then slacks the extreme value for each leg's friction constraint.

Hexapod robots often use a tripod or quadruped gait for locomotion. That is, $N_s = 3$ or $N_s = 4$. When $N_s = 3$, the vector Δ needs to simultaneously satisfy the $2N_s = 6$ conditions because the vectors $E_1^t A^\dagger, \dots, E_3^t A^\dagger$ and the matrices $E_1^n A^\dagger, \dots, E_3^n A^\dagger$ are different from each other. However, the vector Δ is 6-D, and the maximal number of different conditions that can be satisfied simultaneously is five rather than six or eight. Thus, the items are incompatible.

Theorem 2: If there exists a group of contact forces $f = [f_1^\top, \dots, f_{N_s}^\top]^\top \in \mathbb{R}^{N_s}$ to make (41) true, we will find a group of contact forces f^j to make (39) true

$$\begin{cases} \dot{Q} + w \times Q = F_B + F_G + F_b^f + F_l^f + F_T + A_f f \\ \dot{L} + w \times L + v \times Q = M_{BG} + M_b^f + M_l^f + M_T + A_m f \\ \|B_t f_i\| \leq u^\top f_i + \frac{(s_i - l_i) \cdot V_I}{\sum_{i=1}^{N_s} (s_i - l_i)} \end{cases} \quad (41)$$

where $V_I = l_I - \sum_{i=1}^{N_s} s_i$ is an integrated realizable extreme value, $l_I = -(r_f + r_m) \|E_I^n A^\dagger\|$, $E_I^n = \sum_{i=1}^{N_s} E_i^n \in \mathbb{R}^{1 \times 3N_s}$.

This theorem's proof process is provided in Appendix C.

TABLE II
GEOMETRIC PARAMETERS OF THE ROBOT

l_1	l_2	l_3	l_4
0.38 m	0.2 m	0.3 m	0.45 m
d_1	d_2	r	r_c
0.6 m	0.2 m	8.5 cm	$[0, -0.005, 0]^\top$ m

TABLE III
HYDRODYNAMIC PARAMETERS OF THE ROBOT

λ_1	λ_2	λ_3	λ_4	λ_5
5.7 kg	8.6 kg	6.8 kg	0.19 kg·m ²	0.28 kg·m ²
λ_6	c_x	c_y	c_z	m_x
0.2 kg·m ²	48.2 kg/s	76 kg/s	63.2 kg/s	3.7 kg/s
m_y	m_z	μ_l		
5.6 kg/s	4.4 kg/s	0.1 Ns/(m·rad)		

Compared with the leg-terrain fraction constraint in (40), we can obtain a looser friction constraint for i th leg as follows:

$$\|B_t f_i\| \leq u^\top f_i + \frac{(s_i - l_i) \cdot V_I}{\sum_{i=1}^{N_s} (s_i - l_i)}. \quad (42)$$

Based on the established model in (33) and the formulated tight friction constraint in (42), we use a quadratic programming algorithm to find the feasible force vector ξ that is aligned with the predesigned direction ϑ_j as follows:

$$\begin{aligned} & \max \vartheta_j^\top \xi \\ & \text{s.t. (33), (42), and } u^\top f_i \leq \Gamma_i. \end{aligned} \quad (43)$$

Then, the 6-D feasible region can be approximated as a convex set $\mathcal{A}_n$, which is represented as

$$\mathcal{A}_n = \left\{ \sum_{j=1}^{N_\theta} \beta_j \xi_j^* \mid \beta_1 \geq 0, \dots, \beta_{N_\theta} \geq 0, \sum_{j=1}^{N_\theta} \beta_j = 1 \right\} \quad (44)$$

where β_j is the j th coefficient, ξ_j^* is the extremal vertex along the j th search direction.

The methods of checking and enhancing the robot's stability are the same as the approaches in Section V-B.

VI. SIMULATION

We have developed a rough terrain with 6 m in length and 2 m in width in the robot simulation environment(Gazebo). The height function of the terrain is set as $y = \sum_{i=0}^8 p_i x^i$, where $p_0 = -5$, $p_1 = 0.386$, $p_2 = -2.07$, $p_3 = 3.554$, $p_4 = -2.211$, $p_5 = 0.624$, $p_6 = 0.081$, $p_7 = 0.00385$, $p_8 = 1.646 \times 10^{-5}$.

We have also developed a virtual robot in Gazebo, which shares the same geometric and hydrodynamic parameters as the real robot. The parameters are provided in Tables II and III. The mass and buoyancy can be changed in Gazebo according to different simulation objectives. To verify the effectiveness of the proposed criterion and stability enhancement method, we let our robot walk on the surface of the rough terrain and carry out three groups of simulations.

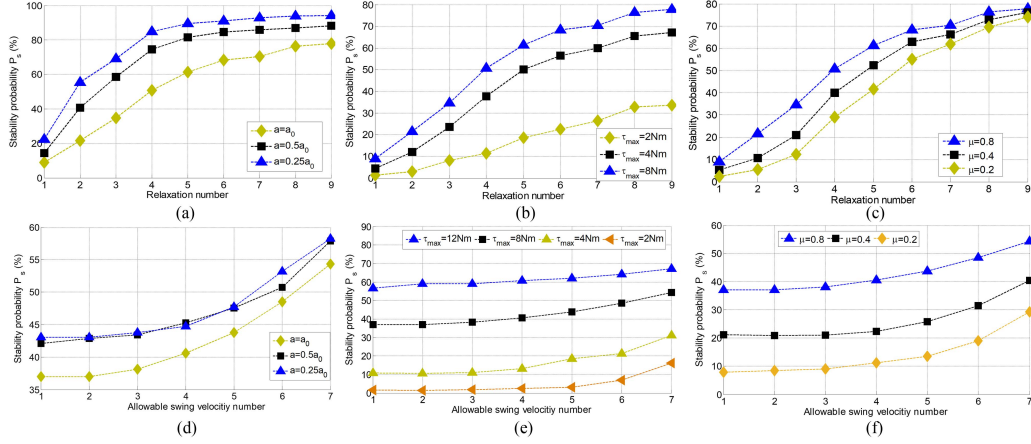


Fig. 9. Stability probability of robot climbing rough terrain under different relaxations, torque limits, friction coefficients, and accelerations.

A. Stability Analysis With Different Relaxations, Accelerations, Torque Limits, and Friction Coefficients

This section aims to observe how different relaxations, accelerations, torque limits, and friction coefficient affect the robot's stability probability. The robot's mass and buoyancy are set as 15.63 kg and 151.2 N, and the vertical thrust is set as $F_{Ty} = -40$ N, $F_{Tx} = F_{Tz} = M_{Tx} = M_{Ty} = M_{Tz} = 0$. The robot is commanded to climb the terrain with a tripod gait. Then, we collect the corresponding motion data of one climbing process.

To verify the effectiveness of the vertical thrust based stability criterion in Section IV, we set nine types of relaxations: $\Omega_{F1} = [0, 0, 0]^T$, $\Omega_{M1} = [0, 0, 0]^T$, $\Omega_{F2} = [5, 0, 5]^T$, $\Omega_{M2} = [1, 0, 1]^T$, $\Omega_{F3} = [10, 0, 10]^T$, $\Omega_{M3} = [2, 0, 2]^T$, $\Omega_{F4} = [20, 0, 20]^T$, $\Omega_{M4} = [4, 0, 4]^T$, $\Omega_{F5} = [30, 0, 30]^T$, $\Omega_{M5} = [6, 0, 6]^T$, $\Omega_{F6} = [40, 0, 40]^T$, $\Omega_{M6} = [8, 0, 8]^T$, $\Omega_{F7} = [50, 0, 50]^T$, $\Omega_{M7} = [10, 0, 10]^T$, $\Omega_{F8} = [70, 0, 70]^T$, $\Omega_{M8} = [14, 0, 14]^T$, $\Omega_{F9} = [100, 0, 100]^T$, $\Omega_{M9} = [20, 0, 20]^T$. In addition, we decrease the accelerations of the collected data to its 0.5 and 0.25 to set three kinds of accelerations. That said, they are set as $a = a_0$, $a = 0.5a_0$, and $a = 0.25a_0$, where a_0 is the original acceleration. We set four types of torque limits, e.g., $\tau_{\max} = 12 \text{ N} \cdot \text{m}$, $\tau_{\max} = 8 \text{ N} \cdot \text{m}$, $\tau_{\max} = 4 \text{ N} \cdot \text{m}$, and $\tau_{\max} = 2 \text{ N} \cdot \text{m}$. Finally, the friction coefficients are set as four types, e.g., $\mu = 0.8$, $\mu = 0.4$, $\mu = 0.2$, and $\mu = 0.1$.

In addition, to verify the effectiveness of the 6-D thrust based stability criterion in Section V, we set seven groups of allowable swing velocities $\Theta_{\min}$, $\Theta_{\max}$ and $\Upsilon_{\min}$, $\Upsilon_{\max}$. Specifically, $\Theta_{\max} = -\Theta_{\min} = \Upsilon_{\max} = -\Upsilon_{\min} = 9.6, 4.8, 2.4, 1.2, 0.6, 0.3$, or 0.01 rad/s . If the allowable swing velocity is set as 9.6 rad/s , we regard this as the seventh group. We regard it as the first group if it is set as 0.01 rad/s . $\chi_1 = 0.5$, $\chi_2 = 0.5$. The set of accelerations, torque limits, and friction coefficients is the same as the section of verifying the vertical thrust-based stability criterion. We set 76 search directions, which are described as

$$\underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \end{bmatrix}}_{\text{Along the axis } \vartheta_1^T, \dots, \vartheta_{12}^T}, \underbrace{\begin{bmatrix} 0.4082 \\ 0.4082 \\ 0.4082 \\ 0.4082 \\ 0.4082 \\ 0.4082 \end{bmatrix}, \dots, \begin{bmatrix} -0.4082 \\ -0.4082 \\ -0.4082 \\ -0.4082 \\ -0.4082 \\ -0.4082 \end{bmatrix}}_{\text{Between the axis } \vartheta_{13}^T, \dots, \vartheta_{76}^T}.$$

Fig. 9(a)–(c) shows stability probabilities under the vertical thrust-based stability criterion, where the probability P_s equals the count of being stable divided by the total count of the climbing process. Fig. 9(a) shows that the greater relaxation results in the larger stability probability. When the acceleration reduces, the probability will be increased. Fig. 9(b) demonstrates that a smaller torque limit can result in a smaller stability probability because it will enhance the probability of legs collapsing. Fig. 9(c) shows that the stability probability gets smaller when the friction coefficient gets smaller. The probability differences caused by friction differences in the first few groups of relaxations are much larger than in the latter groups. The phenomenon arises because the friction coefficient is the dominant factor in determining the probability when relaxations are small. Fig. 9(d)–(f) provides the stability probabilities under the 6-D thrust based stability criterion with different accelerations, torque limits, and friction coefficients, respectively. The change of the stability probabilities response to accelerations, torque limits, and friction coefficients are similar to Fig. 9(a)–(c). Compared with the relaxations in vertical thrust-based criterion, the deduce of the allowable swing velocity cannot make the probability drop drastically because it is not the only factor to determine the F_{low} , F_{up} , M_{low} , and M_{up} . The magnitude of stability probabilities in Fig. 9 coincides with the actual experiences and verifies the two criteria' effectiveness.

Fig. 10 presents the curves of the maximum vertical thrust $F_{Ty}^{\min}$, minimum thrust $F_{Ty}^{\max}$, and robot's thrust F_{Ty} under the seventh relaxation. The trends of these curves contribute to clarifying how the factors, such as acceleration, maximum torque, and friction coefficient, affect the probability. Fig. 10(a)–(c) presents the $F_{Ty}^{\min}$, $F_{Ty}^{\max}$, and F_{Ty} with three kinds of accelerations. We observe that smaller acceleration can result in a larger maximum vertical thrust and smaller minimum thrust, which contributes to forming a larger region of allowable thrusts because it does not need to suppress a force that is too large for inertia under the prescriptive constraints. Fig. 10(c)–(f) provides the thrust curves under the torque limits of $8 \text{ N} \cdot \text{m}$, $12 \text{ N} \cdot \text{m}$, $4 \text{ N} \cdot \text{m}$, and $2 \text{ N} \cdot \text{m}$, respectively. We find that the smaller torque limit can enlarge the value of the minimum thrust and reduce the probability. Furthermore, Fig. 10(c) and (d) shows that the increase of the limit cannot change the value of $F_{Ty}^{\min}$ and $F_{Ty}^{\max}$ significantly because their limits are large enough. Fig. 10(c) and (g)–(i) provides the $F_{Ty}^{\min}$, $F_{Ty}^{\max}$, and F_{Ty} with four types

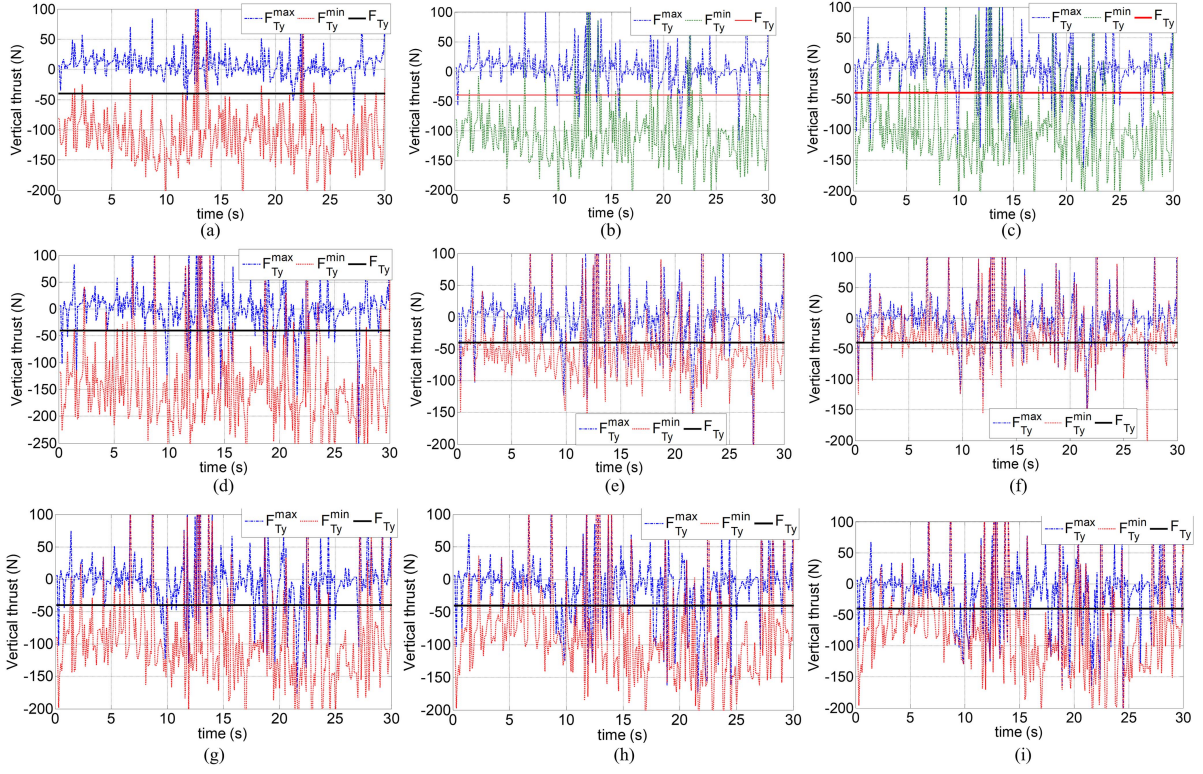


Fig. 10. Maximum, minimum, and robot's vertical thrust of seventh relaxation with different accelerations, maximum torques, and friction coefficients. (a) $a = 0.25a_0$, $\tau_{\max} = 8 \text{ N} \cdot \text{m}$, $\mu = 0.8$. (b) $a = 0.5a_0$, $\tau_{\max} = 8 \text{ N} \cdot \text{m}$, $\mu = 0.8$. (c) $a = a_0$, $\tau_{\max} = 8 \text{ N} \cdot \text{m}$, $\mu = 0.8$. (d) $a = a_0$, $\tau_{\max} = 12 \text{ N} \cdot \text{m}$, $\mu = 0.8$. (e) $a = a_0$, $\tau_{\max} = 4 \text{ N} \cdot \text{m}$, $\mu = 0.8$. (f) $a = a_0$, $\tau_{\max} = 2 \text{ N} \cdot \text{m}$, $\mu = 0.8$. (g) $a = a_0$, $\tau_{\max} = 8 \text{ N} \cdot \text{m}$, $\mu = 0.4$. (h) $a = a_0$, $\tau_{\max} = 8 \text{ N} \cdot \text{m}$, $\mu = 0.2$. (i) $a = a_0$, $\tau_{\max} = 8 \text{ N} \cdot \text{m}$, $\mu = 0.1$.

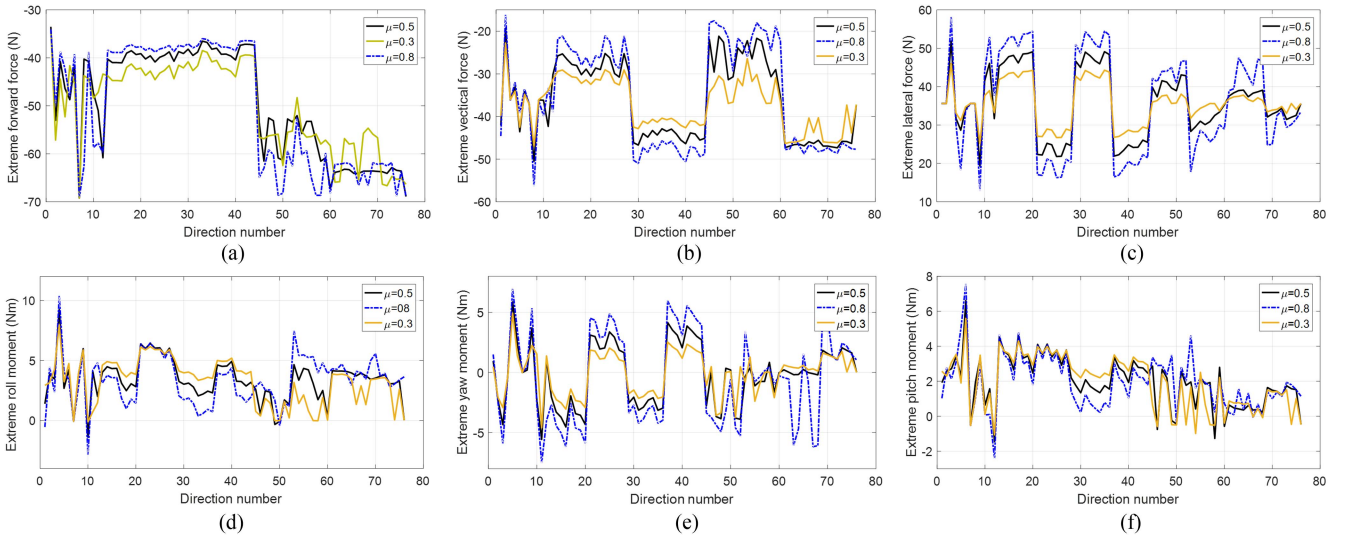


Fig. 11. Extremal thruster forces along the set 76 search directions with three different friction coefficients.

of friction coefficients, e.g., $\mu = 0.8$, $\mu = 0.4$, $\mu = 0.2$, and $\mu = 0.1$.

The relaxations and accelerations are related to the smoothness of gait planning because nonsmooth gaits can lead to large leg-terrain interaction forces and a poor leg alternating character. In addition, the friction coefficient is related to the leg-terrain mechanics. Finally, the torque limit is determined by the chosen driving motor in the hip joint.

Figs. 11–13 present one motion state's extremal vertex (extremal and feasible 6-D thruster force) along the designed 76 search directions with different friction coefficients, torque limits, and upper bounds of unmodeled dynamics. Figs. 11 and 12 show that a smaller friction coefficient and torque limit result in a narrower convex set (stable region). That said, it raises the difficulty of finding a stable thrust vector. Fig. 13 shows that larger upper bounds of unmodeled dynamics can

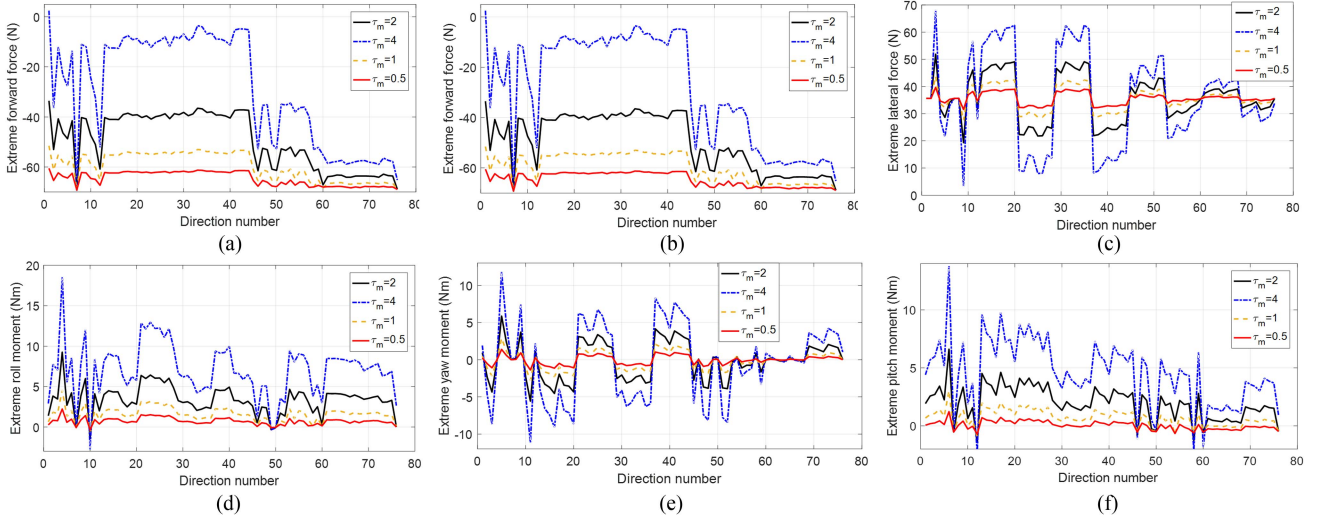


Fig. 12. Extremal thruster force along the set 76 search directions with four different torque limits.

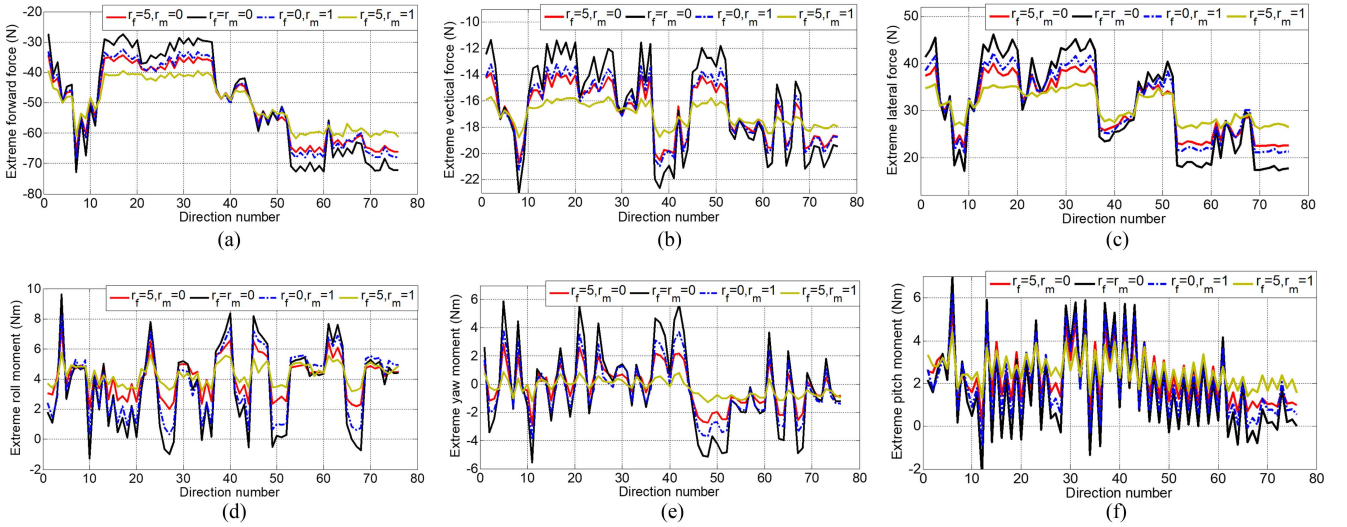


Fig. 13. Extremal thruster force along the set 76 search directions with four groups of upper bounds of unmodeled dynamics.

lead to a narrower stable region. Thus, the established force balancing model should be accurate enough to accurately judge and effectively enhance the robot's stability.

B. Stability Analysis With Different Robot Masses

In this section, we conduct two groups of simulations under four kinds of masses and buoyancies, where the four masses are set as 67 kg, 37 kg, 15.63 kg, and 5 kg, and the corresponding buoyancy is 2 N larger than the gravity. The torque limit and friction coefficient are set as $\tau_{\max} = 2.5 \text{ N} \cdot \text{m}$ and $\mu = 0.8$. We use the two groups of simulations to observe the stability probability of the two criteria, and each group has four climbing tests. We collect the motion states of eight climbing tests and check the robot's stability by using the proposed two criteria. In vertical thrust based criterion, the relaxation is set as $\{\Omega_{F5}, \Omega_{M5}\}$. In the 6-D thrust-based criterion, the allowable swing velocities are selected as $\Theta_{\max} = -\Theta_{\min} = \Upsilon_{\max} = -\Upsilon_{\min} = 4.8 \text{ rad/s}$.

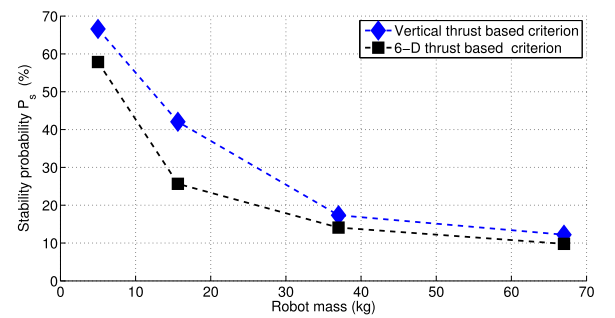


Fig. 14. Stability probability under four different masses and relaxations.

Fig. 14 shows the stability probability of the four tests with different masses. When the mass is larger, the stability probability of the robot is smaller, and the robot can be more unstable. The

TABLE IV
STABILITY PROBABILITIES OF 12 GROUPS

Group ID	Probability in Test 1[%]	Probability in Test 2[%]	Probability in Test 3[%]	Probability in Test 4[%]	Probability in Test 5[%]	Average probability
1	50.34	51.62	51.36	47.78	52.05	50.63
2	61.47	56.72	58.53	54.37	54.83	57.18
3	66.59	64.68	54.36	67.88	70.29	64.76
4	38.99	36.54	44.11	24.67	41.26	37.11
5	75.73	81.58	78.63	81.26	80.28	79.50
6	86.71	85.17	85.05	83.37	84.71	85.00
7	31.32	29.90	27.77	28.41	25.74	35.79
8	48.33	50.35	44.58	49.25	47.62	48.03
9	54.71	57.85	55.83	55.56	53.03	55.40
10	33.43	27.91	26.76	24.30	31.78	28.84
11	53.65	52.74	60.62	60.35	52.80	56.03
12	66.51	66.26	65.19	61.15	67.59	66.34

large mass results in a large inertia force and may hinder the robot from suppressing it. The magnitude of stability probabilities in Fig. 14 coincides with the robot's swing and sliding degree, which verifies the two criteria' effectiveness. In addition, our stability criteria provide a necessary condition that combines the torque limit and the friction coefficient with the used thrust to design the robot mass for stable locomotion.

C. Stability Enhancement

To verify the effectiveness of the stability enhancement method, we conduct 12 groups of simulations, where the robot's mass and buoyancy are set as 5 kg and 50 N, and each group consists of five climbing tests. The vertical thrusts of the groups 1 and 7, 2 and 8, 3 and 9, 4 and 10 are set as -8 , -20 , -40 , and -80 N, respectively. The torque limit and friction coefficient are set as $\tau_{\max} = 2.5 \text{ N} \cdot \text{m}$ and $\mu = 0.8$.

We use the vertical thrust-based criterion and stability enhancement method to execute the first six groups of simulations. Then, we use the 6-D thrust based criterion and method to perform the latter six groups. In vertical thrust based criterion, we use the fifth relaxation, i.e., $\{\Omega_{F5}, \Omega_{M5}\}$. The vertical thrusts of the latter two groups vary according to the calculated maximum and minimum forces. In the 6-D thrust based criterion, the allowable swing velocities are selected as $\Theta_{\max} = -\Theta_{\min} = \Upsilon_{\max} = -\Upsilon_{\min} = 4.8 \text{ rad/s}$. Specifically, in the fifth group, the vertical thrust F_{Ty} is designed as $F_{Ty} = 0.5(F_{Ty}^{\min} + F_{Ty}^{\max})$, which aims to force the robot owning largest stability margin. In the sixth group, the vertical thrust F_{Ty} is also equal to the one at the previous moment if the robot is stable. Otherwise, F_{Ty} is designed to make the robot in a critical steady state. That said, it aims to make $\Psi = 0$ when the robot is unstable. Therefore, the vertical thrust is set as

$$F_{Ty} = \begin{cases} F_{Ty}^{\min}, & \text{if } |F_{Ty}^{\min} - F'_{Ty}| < |F_{Ty}^{\max} - F'_{Ty}| \\ F_{Ty}^{\max}, & \text{else} \end{cases}$$

where F'_{Ty} is the vertical thrust of the previous moment. Similarly, the 11th and 12th groups use the stability enhancement methods in (37) and (38) to make the robot in a most steady state and critical steady state.

Table IV provides the stability probabilities of 12 groups of simulations. The average probabilities of the first four groups are 50.63%, 57.18%, 64.76%, and 37.11%, respectively. The average probability of group 4 is much less than groups 1, 2,

and 3. The average probabilities of groups 5 and 6 are 79.50% and 85.00%, respectively, which are larger than those of the first four groups, and the probability of group 5 is smaller than that of group 6. Fig. 15 provides the robot's vertical thrust and the maximum/minimum vertical thrusts of the first test of each group. We find that the frequency of vertical thrust belonging to the region formed by maximum/minimum thrusts in the fifth and sixth groups is higher than in the first four groups. The mutual magnitude character of stability probabilities between the latter six groups is similar to the first six groups.

Based on the abovementioned results, we obtain the following findings.

- 1) The average probabilities in group 4 show that the robot is probably unstable if the vertical thrust is too large. Specifically, the robot may fall because the torque limit is small, and the hip joint cannot provide enough torque.
- 2) Compared with the first four groups, the robot in the fifth and sixth groups has better stability, which verifies the effectiveness of the vertical thrust-based stability enhancement method. In addition, we observe that the stability worsens if the robot climbs with the maximum stability margin because it will lose part of its maneuverability, making it difficult to track the desired motion and reduce its stability flexibly.
- 3) The stability probabilities of ten tests in groups 5 and 6 are less than 86%, which cannot be improved to 100% because the robot's stability is enhanced by only relying on the vertical thrust. Concretely, the large leg-terrain interaction force can result in a large inertial force. Depending on the vertical thrust may not suppress the force, leading to the robot's instability. Reasonable planning the gait is a solution to improve the stability probability, but it cannot be 100%.
- 4) Theoretically, the 6-D thrust-based stability enhancement method can ensure the robot's one hundred percent stability by taking full advantage of 6-D thruster forces and torques. However, it depends on two preconditions: 1) the established model should be accurate enough; 2) the allowable swing velocities should match the robot's motion planner enough. Once they are inaccurate or match, the 6-D thrust based method will result in inaccurate extremal thruster forces. The established model in this article is based on four assumptions. Thus, the stability probability in group 12 cannot be 100%.

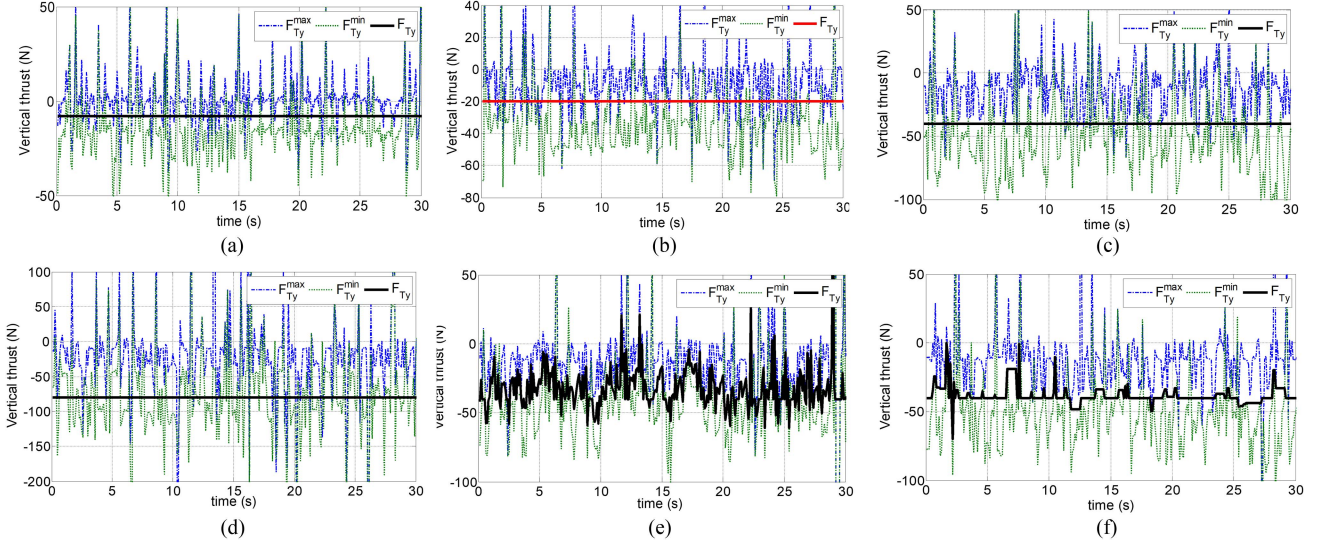


Fig. 15. Stability probability of six different tests. (a) Test 1 in group 1. (b) Test 1 in group 2. (c) Test 1 in group 3. (d) Test 1 in group 4. (e) Test 1 in group 5. (f) Test 1 in group 6.

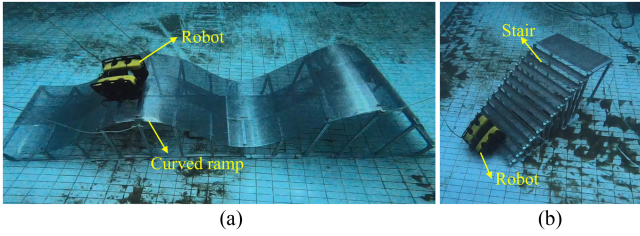


Fig. 16. Experimental scenario. (a) Curved ramp. (b) Stair.

VII. EXPERIMENTS

A. Experiment Setup

In this section, we experiment to evaluate the proposed stability criterion and its enhancement method. We present the experimental scenario in Fig. 16(a), where the robot is on the surface of a man-made rough terrain (curved ramp) and the terrain is on the bottom of a 10 m×10 m×8 m pool. The terrain consists of aluminum alloy skeletons and iron sheets; its length and width are 6 m and 2 m, respectively. The height function of the man-made terrain is similar to the one in the simulation, and it can be described as $y = \sum_{i=0}^8 p_i x^i$, where $p_0 = -7.47$ and the parameters $p_1, \dots, p_8$ are the same ones in simulation. In addition, the robot's geometric and hydrodynamic parameters are shown in Tables II and III. As shown in Fig. 16(b), we construct a stair using the aluminium alloy skeletons and iron sheets and then present it on the bottom of the pool. We experiment climbing a stair in an indoor pool, where the stair has 12 steps, each with a length 0.15 m, width 1.2 m, and height 0.1 m.

In the experiment of climbing the curved ramp, the robot's mass and buoyancy are 37 kg and 364 N. The friction coefficient is $\mu = 0.6$. The torque limit is $\tau_{\max} = 16 \text{ N} \cdot \text{m}$, and the relaxation is set as $\Omega_F = [50, 0, 50]^T$, $\Omega_M = [10, 0, 10]^T$. Similar to the settings in Section VI-C, we conduct 12 experimental groups to verify the effectiveness of the vertical thrust based and

6-D thrust based criteria and stability enhancement methods, each consisting of five climbing tests. The vertical thrusts of the groups 1 and 7, 2 and 8, 3 and 9, 4 and 10 are set as -8 , -20 , -40 , and -80 N , respectively. The method of calculating the fifth, sixth, 11th, and 12th groups' thrusts is presented in Section VI-C. In the experiment of climbing the stairs, the robot mass, buoyancy, friction coefficient, torque limit, and relaxation are the same as climbing the curved ramp. We use the method in group 6 to check whether our stability enhancement method could make the robot climb more challenging rough terrain.

B. Experiment Results

Table V provides the stability probability of six groups of experiments. The average probabilities of the first four groups are 32.55%, 36.96%, 41.26%, and 28.22%, respectively. The average probabilities of groups 1 and 4 are less than those in groups 2 and 3. The average probabilities of groups 5 and 6 are 50.73% and 52.03%, respectively, which are larger than those of the first four groups, and the probability of group 5 is smaller than that of group 6. Fig. 17 presents six snapshots of each group's Test 1. The climbing phenomenons of each test are consistent with the probabilities in Table IV. Specifically, the robot in group one easily slipped and could not complete the climbing task because using the small vertical thruster force. The large vertical force in group 4 can lead to the robot's shaking, although the robot can finish climbing. The robot in the other groups can climb the terrain smoothly using a relatively reasonable vertical force. Similar to Section VI-C, the experimental results in groups 7–12 are performed by the 6-D thrust-based stability enhancement method. Fig. 18 shows that the vertical thrust-based stability enhancement method can make the robot climb the stair successfully, and its stability probability is 50.27%. We can obtain the appropriate constant vertical thrust to successfully make the robot climb the stairs. However, many trials are needed to find the constant thrust. The stability enhancement method can overcome this issue by automatically adjusting based on the judged stability.

TABLE V
STABILITY PROBABILITY OF 12 GROUPS OF EXPERIMENTS

Group ID	Probability in Test 1[%]	Probability in Test 2[%]	Probability in Test 3[%]	Probability in Test 4[%]	Probability in Test 5[%]	Average probability
1	21.58	41.18	31.1	32.71	36.19	32.55
2	40.46	36.41	35.35	44.92	27.68	36.96
3	53.03	37.07	34.22	41.16	40.81	41.26
4	29.9	25.74	29.12	28.26	28.22	28.22
5	53.18	55.56	47.37	48.70	48.85	50.73
6	57.3	44.03	47.06	53.87	57.88	52.03
7	27.32	25.79	35.35	28.77	26.89	28.82
8	34.89	35.49	29.68	36.29	31.26	33.52
9	45.28	40.56	39.41	36.57	45.26	41.41
10	24.35	29.38	30.58	29.27	25.48	27.81
11	56.32	52.37	50.24	46.29	46.51	50.35
12	55.30	50.26	56.34	50.28	53.29	53.09

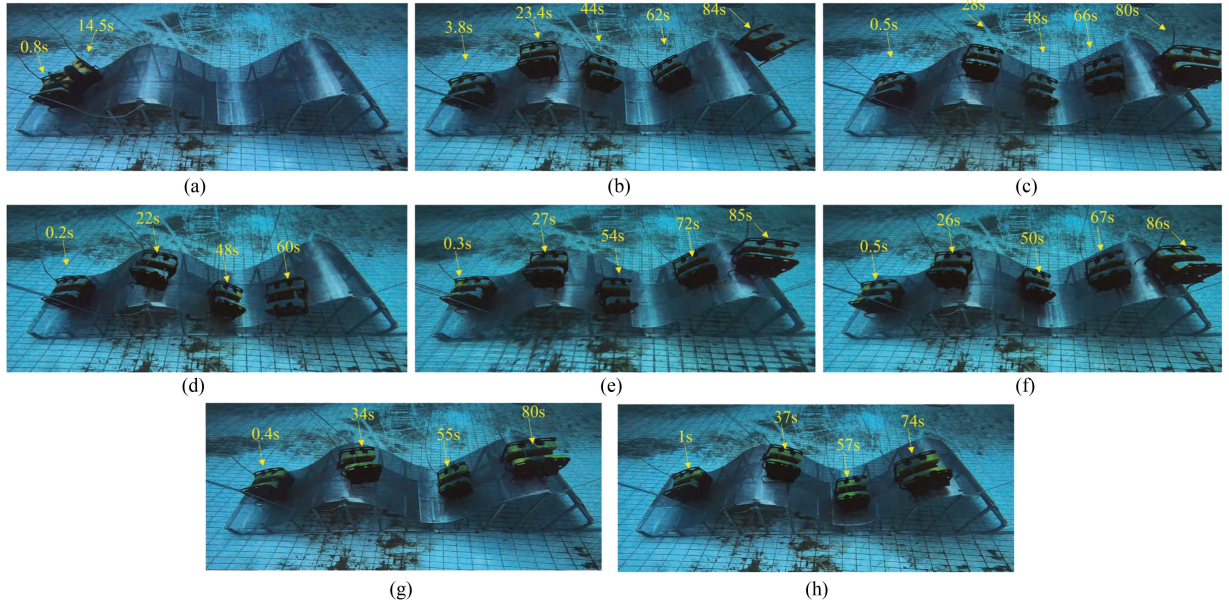


Fig. 17. Snapshot of climbing a rough terrain. (a) Test 1 in group 1, where the robot cannot walk due to legs' skidding. (b) Test 1 in group 2. (c) Test 1 in group 3. (d) Test 1 in group 4, where the robot cannot complete the climbing task. (e) Test 1 in group 5. (f) Test 1 in group 6. (g) Test 1 in group 11. (h) Test 1 in group 12.

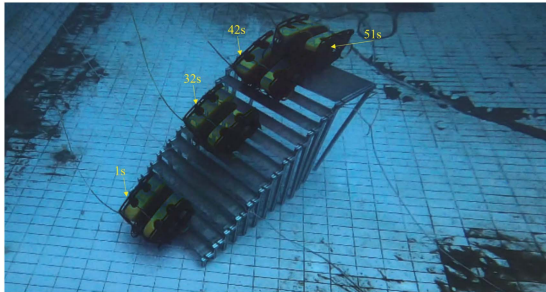


Fig. 18. Snapshot of climbing a stair.

C. Discussion

We find the results between simulations in Section VI-C and experiments are different, and they arise from the differences in the chosen parameters, such as the robot's mass, friction

coefficient, and torque limit. Then, we will introduce the two reasons accounting for the parameters' difference.

- 1) In the experiment, the rough terrain's surface is made of an iron sheet, which is only three millimetres thick and can deform slightly under pressure. The buffer caused by the deformation can reduce the leg-terrain impact forces. The simulation's terrain is hard and can result in larger leg-terrain impact forces. Then, in the simulation, we reduce the robot's mass to 5 kg because the smaller mass can result in smaller impact forces, generating smaller inertia forces that can be suppressed easily.
- 2) Since the mass in the simulation is small, we only need to use much smaller torque limits to steer the legs. Thus, the torque limit in simulation is set as $2.5 \text{ N} \cdot \text{m}$, which is much smaller than the value of $16 \text{ N} \cdot \text{m}$ in the experiment. The application prospects of our methods are list as follows.
 - 1) The presented stability criterion and enhancement method can be expanded to other thruster-assisted legged robots with some modifications. For example, our methods can be

applied to judge and enhance the stability of aerial thruster-assisted legged robots, such as the robot in [41]. We need to remove the fluid forces applied to the robot's body and legs and modify the contact matrices in Section II-D because its legs are not C-shaped. In addition, our criterion can also be used to judge the stability of wheeled robots but not tracked robots because the wheeled robots touch the terrain with a discrete support manner. We only need to rederive the contact matrices according to the specific form-position relationship between the wheels' touching points and the robot's CoM in Section II-D.

- 2) Our stability criteria can be used as a constraint while designing the gait planner and controller. In addition, we can use our stability enhancement method to obtain the desired stability margin by suppressing the external disturbances and model uncertainties.
- 3) Our stability criteria relate to robot mass, torque limits of driving joints, leg-terrain friction coefficient, legs' shape, and thruster forces. Then, our criterion can be applied to guide the design of the robot's design parameters, such as robot mass and geometric parameters. It can also be used to guide the choice of joints' motors and thrusters to obtain the rational torque limits and thrusts and achieve the desired locomotion.
- 4) Our stability enhancement method provides a solution to maintain the robot's stability by regulating the thruster forces and joint torques separately.

VIII. CONCLUSION

In this article, we have presented a new method to define the three types of walking motion under the established force balancing model. In addition, we have proposed two criteria to judge the stability of our thruster-assisted underwater hexapod robots by checking whether the current thrust belongs to the allowable region, where the region is calculated by the optimization method. We have verified our criterion's effectiveness and clarified how factors, such as relaxations, maximum torques, and friction coefficients, affect the stability through multiple comparative simulations and experiments. On this basis, we have presented a method to enhance stability by changing the value of the vertical thrust. Simulation and experiment results have demonstrated that our method enhances the robot's stability probability by 15%–20%.

APPENDIX A PROOF OF LEMMA 1

Taking the partial derivative of the first line of (30) with respect to $\dot{q}_d$, we have

$$\begin{aligned} \frac{\partial u_1}{\partial \dot{q}_d} &= -ac_1 \left(\frac{1}{2} \right)^{\chi_1} \left(-2\chi_1((q - q_d)^2 + (\dot{q} - \dot{q}_d)^2)^{\chi_1-1} \right. \\ &\quad \left. + \frac{((q - q_d)^2 + (\dot{q} - \dot{q}_d)^2)^{\chi_1}}{(\dot{q} - \dot{q}_d)^2} \right) \\ &= ac_1 \left(\frac{1}{2} \right)^{\chi_1} [(q - q_d)^2 + (\dot{q} - \dot{q}_d)^2]^{\chi_1} \\ &\quad \cdot \left[\frac{2\chi_1}{(q - q_d)^2 + (\dot{q} - \dot{q}_d)^2} - \frac{1}{(\dot{q} - \dot{q}_d)^2} \right] \end{aligned}$$

$$= M [(2\chi_1 - 1)(\dot{q} - \dot{q}_d)^2 - (q - q_d)^2] \quad (45)$$

where $M = ac_1 \left(\frac{1}{2} \right)^{\chi_1} \cdot \frac{[(q - q_d)^2 + (\dot{q} - \dot{q}_d)^2]^{\chi_1-1}}{(\dot{q} - \dot{q}_d)^2} > 0$.

Since $\chi_1 \in (0, 0.5]$, we have $2\chi_1 - 1 \leq 0$. Then, we can derive that $\frac{\partial u_1}{\partial \dot{q}_d} < 0$. That said, the function u_1 in (30) is monotonically decreasing in the variable of $\dot{q}_d$. Similarly, we have $\frac{\partial u_2}{\partial \dot{\zeta}_d} < 0$.

APPENDIX B PROOF OF THEOREM 1

Proof: Based on (1) and (39), we have

$$Af = Af^j + \Delta \quad (46)$$

where $\Delta = [\Delta F_j^\top, \Delta M_j^\top]^\top \in \mathbb{R}^6$.

Then, we have $f^j = f - A^\dagger \Delta$, $u^\top f_i$ in (1) and $u^\top f_i^j$ in (39) can be rewritten as $E_i^n f$ and $E_i^n f^j$, respectively. Then, we have

$$\begin{aligned} u^\top f_i^j - u^\top f_i &= E_i^n (f^j - f) = -E_i^n A^\dagger \Delta \\ &\geq -\|E_i^n A^\dagger\| \cdot \|\Delta\| \geq -(r_f + r_m) \|E_i^n A^\dagger\| \geq l_i. \end{aligned} \quad (47)$$

Similarly, the $B_t f_i$ in (1) and $B_t f_i^j$ in (39) can be rewritten as $E_i^t f$ and $E_i^t f^j$, respectively. Then, we have

$$\begin{aligned} \|B_t f_i^j\| &= \|E_i^t f^j\| = \|E_i^t (f - A^\dagger \Delta)\| \\ &\leq \|E_i^t f\| + \|E_i^t A^\dagger \Delta\| \leq \|E_i^t f\| + \bar{\sigma}(E_i^t A^\dagger) \|\Delta\| \\ &\leq \|E_i^t f\| + (r_f + r_m) \bar{\sigma}(E_i^t A^\dagger) \leq \|B_t f_i\| + s_i \end{aligned} \quad (48)$$

where $\bar{\sigma}(\cdot)$ is the largest singular value of the matrix $(\cdot)$.

Based on the inequations in (47) and (48), we can use $\|B_t f_i\| \leq u^\top f_i + l_i - s_i$ to deduce that $\forall \Delta F_j, \Delta M_j, \|\Delta F_j\| < r_f, \|\Delta M_j\| < r_m, \|B_t f_i^j\| \leq \|B_t f_i\| + s_i \leq u^\top f_i + l_i \leq u^\top f_i^j$.

APPENDIX C PROOF OF THEOREM 2

Proof: Summing up N_s inequations where i th inequation is shown in (48), we have $\sum_{i=1}^{N_s} (\|B_t f_i^j\| - \|B_t f_i\|) \leq \sum_{i=1}^{N_s} s_i$. Referring to the deducing method in (48), we have

$$\begin{aligned} \sum_{i=1}^{N_s} u^\top (f_i^j - f_i) &= E_I^n (f^j - f) = -E_I^n A^\dagger \Delta \\ &\geq -(r_f + r_m) \|E_I^n A^\dagger\|. \end{aligned} \quad (49)$$

Thus, we can use $\sum_{i=1}^{N_s} \|B_t f_i\| \leq \sum_{i=1}^{N_s} u^\top f_i + V_I$ to deduce that $\forall \Delta F_j, \Delta M_j, \|\Delta F_j\| < r_f, \|\Delta M_j\| < r_m, \sum_{i=1}^{N_s} \|B_t f_i^j\| \leq \sum_{i=1}^{N_s} \|B_t f_i\| + \sum_{i=1}^{N_s} s_i \leq \sum_{i=1}^{N_s} u^\top f_i + l_I \leq \sum_{i=1}^{N_s} u^\top f_i^j$, where $V_I = l_I - \sum_{i=1}^{N_s} s_i$. Similarly, the priori condition to achieve the extreme value V_I is that the following $N_s + 1$ conditions need to be satisfied simultaneously:

- 1) Δ is perpendicular to the vectors of $E_I^n A^\dagger$;
- 2) Δ is collinear to the max singular vectors of $E_1^t A^\dagger, \dots, E_{N_s}^t A^\dagger$. The hexapod robot in this article walks

by a tripod or quadruped gait. That is, $N_s + 1 = 4$ or 5 , which is less than or equal to the maximal provided number 5 . Thus, the integrated extreme value V_I is realizable.

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Lepeng Chen received the B.Eng. and M.S. degrees in automatic control from Northwestern Polytechnical University, Xi'an, China, in 2015 and 2018, respectively. He is currently working toward the Ph.D. degree in automatic control with the School of Marine Science and Technology, Northwestern Polytechnical University.

His research interests are underwater legged robotics, adaptive control, and motion planning of marine vehicles.



Rongxin Cui (Member, IEEE) received the B.Eng. degree in automatic control and the Ph.D. degree in control science and engineering from Northwestern Polytechnical University, Xi'an, China, in 2003 and 2008, respectively.

He is currently a Professor with the School of Marine Science and Technology, Northwestern Polytechnical University. His current research interests include control and navigation for underwater vehicles.

Dr. Cui is currently an Associate Editor for IEEE TRANSACTIONS ON SYSTEMS, MAN, AND CYBERNETICS: SYSTEMS, and IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS.



Zhijun Li (Fellow, IEEE) received the Ph.D. degree in mechatronics from Shanghai Jiao Tong University, Shanghai, China, in 2002.

He is currently a Chair Professor with Tongji University, Shanghai. He has authored or coauthored more than 400 papers, where the prestigious contributions were wearable robotics and bio-mechatronics systems. His current research interests include wearable robotics, bio-mechatronics systems, intelligent control.

Dr. Li was the recipient of the Distinguished Lecturer (RAS), the Web of Science Highly Cited Researcher (2019–2023), the 2018 National Tenthousand Talents Program in China, 2016 National Distinguished Young Scholar (NSFC). He is a Fellow of and AAIA.



Weisheng Yan received the B.Eng. degree in automatic control and the Ph.D. degree in navigation, guidance, and control from Northwestern Polytechnical University, Xi'an, China, in 1991 and 1999, respectively.

He is currently a Professor with the School of Marine Science and Technology, Northwestern Polytechnical University. His current research interests include guidance, navigation, and control of underwater vehicles.



Hui Xu received the Ph.D. degree in mechanical engineering from Northwestern Polytechnical University, Xi'an, China, in 2006.

He is currently an Associate Professor with the School of Marine Science and Technology, Northwestern Polytechnical University. His research interests include underwater robot designing, bioinspired systems, biomechatronics.



Chenguang Yang (Fellow, IEEE) received the B.Eng. degree in measurement and control from Northwestern Polytechnical University, Xian, China, in 2005 and the Ph.D. degree in control engineering from the National University of Singapore, Singapore, in 2010.

He is currently the Chair of Robotics with the Department of Computer Science, University of Liverpool, Liverpool, U.K. His research interest lies in human–robot interaction and intelligent system design.

Dr. Yang was a recipient of IEEE TRANSACTIONS ON ROBOTICS Best Paper Award in 2012 and IEEE Transactions on Neural Networks and Learning Systems Outstanding Paper Award in 2022, as the lead author.



Haitao Yu received the B.Eng. degree in mechanical engineering from Northwestern Polytechnical University, Xi'an, China, in 2021. He is currently working toward the Ph.D. degree in automatic control with the School of Marine Science and Technology, Northwestern Polytechnical University.

His research interests include mechanical design, biomimetic robotics, and mechatronics.